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Deep learning for computational cytology: A survey

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ABSTRACT

Computational cytology is a critical, rapid-developing, yet challenging topic in medical image computing concerned with analyzing digitized cytology images by computer-aided technologies for cancer screening. Recently, an increasing number of deep learning (DL) approaches have made significant achievements in medical image analysis, leading to boosting publications of cytological studies. In this article, we survey more than 120 publications of DL-based cytology image analysis to investigate the advanced methods and comprehensive applications. We first introduce various deep learning schemes, including fully supervised, weakly supervised, unsupervised, and transfer learning. Then, we systematically summarize public datasets, evaluation metrics, versatile cytology image analysis applications including cell classification, slide-level cancer screening, nuclei or cell detection and segmentation. Finally, we discuss current challenges and potential research directions of computational cytology.

1. Introduction

Cytology (also known as cytopathology) is the examination of cells from body tissues or fluids under microscope, usually for cancer screening and early diagnosis (Gray and Kocjan, 2010; Tan and Tatsumura, 2015). Cells are the structural and functional unit of living organisms, and their appearance and morphology can reflect the biological nature of the organ and even the body (Johnston, 1952; Ji et al., 2020). An illustrative workflow for cytology screening is shown in Fig. 1(A), involving collection and preservation, centrifugation, slide preparation, and staining (Ivanovic, 2014). Cytologists or technicians then observe samples under microscope and examine cells by their morphologies and other properties, looking for abnormal cells (Fig. 1(B)). The samples can also be scanned into whole slide digital images (Fig. 1(C)) for further analysis. In this workflow, the following three types of collection techniques are commonly used to collect cytology specimens: (1) Exfoliative cytology, including sputum, urine sediment, pleural effusion, and ascites (Maharjan et al., 2017). (2) Abrasive cytology, including cervical scraping, gastrointestinal tract, and endoscopic brushing (Burkhardt and Schwarz-Furlan, 2018). (3) Aspiration cytology, also known as fine-needle aspiration cell inspection (FNAC) (Field et al., 2019), usually for breast, thyroid, and lung. Specimen collection in cytology does not require the removal of intact tissue, making it much less invasive and more convenient than other biopsy modalities such as histopathology (Kitchener et al., 2006).

The examination of cells or cell clusters is a highly professional task, requiring formal training and assessments overseen by international bodies. Currently, there is a trend to standardize cytological diagnosis reporting, allowing reports and implied risk of cancers to be understood by clinicians without ambiguity (Barkan et al., 2016). It also provides well-defined features to be looked for among the cells. Fig. 2 illustrates typical cytological morphologies of commonly encountered specimens. Cytology screening was first applied in cervix cancers almost 100 years ago (Shaw, 2000). Current reporting of cervical cytology is guided by the Bethesda system (Nayar and Wilbur, 2015). Cancerous and precancerous cells are first categorized into squamous cells and glandular cells, then they can be identified and graded by combinations of cytological features such as enlarged nuclei, multinucleation, perinuclear halo, increased nuclear to cytoplasm ratio, wrinkled nuclear membrane, dyskeratosis (abnormal keratin formation), prominent nucleoli, and tumor diathesis. Following the success of the Bethesda system, current reporting of breast aspiration cytology is guided by FNAB (Field et al., 2019). Aspiration of malignant breast lesions is often hypercellular and the cancer cells possess large sometimes irregular nuclei with prominent nucleoli and the lack of myoepithelial cells. Different grades of atypia were also established to estimate the risk of malignancy (Beca and Schmitt, 2019). Similarly, bladder cancer cells from urine often show irregular nuclei with a very high N/C ratio (>0.7), prominent

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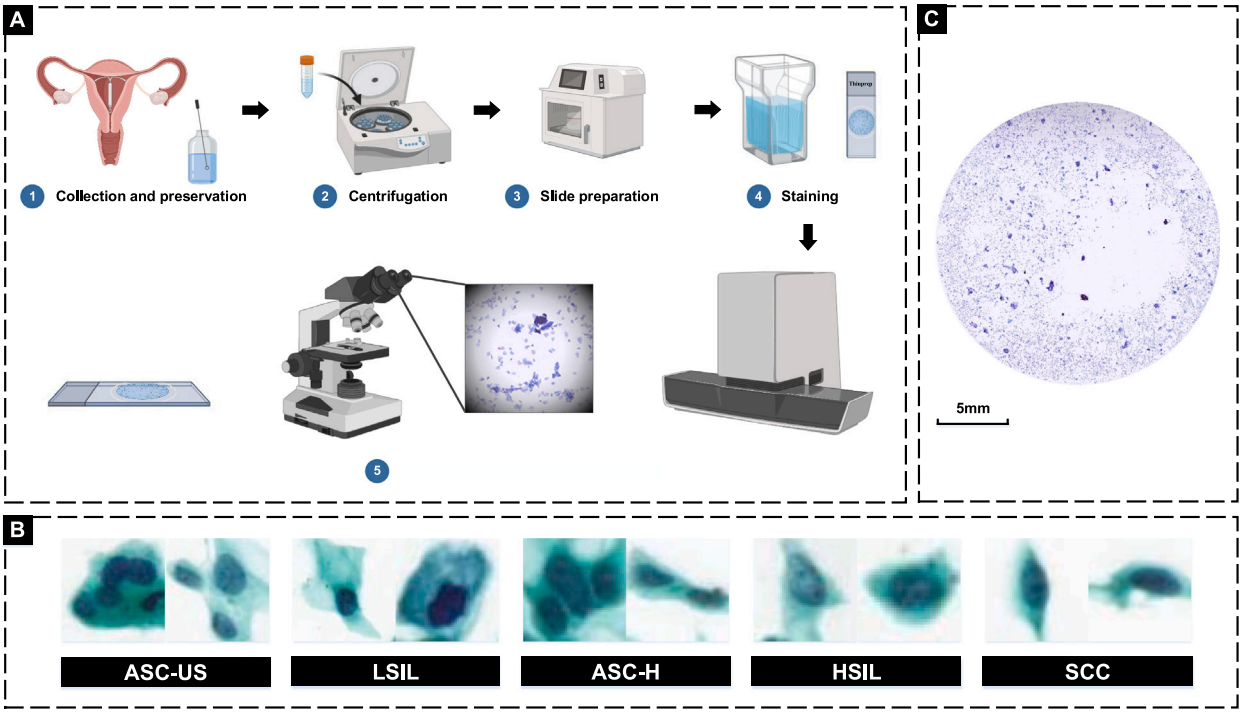

Fig. 1. Illustration of clinical cytology screening. (A) The procedure of cytology specimen preparation (taking the cervix as an example). (B) Cytology images in different categories. (C) Whole slides for digital processing and analysis.

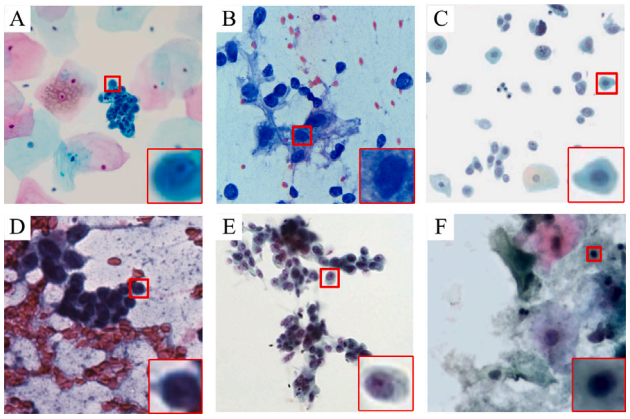

Fig. 2. Typical cytological morphologies of commonly encountered specimens, including (A) Cervix (Zhou et al., 2019b), (B) Breast (Saikia et al., 2019), (C) Urine (Awan et al., 2021), (D) Thyroid (Dov et al., 2021), (E) Lung (Teramoto et al., 2017), (F) Oral (Matias et al., 2021). Lower right red boxes represent zoomed versions of the smaller red boxes in images.

nucleoli and clumped/coarse chromatin, as outlined in Rosenthal et al. (2016). Thyroid cancer aspirates show distinctive features, including papillary structures, psammoma bodies and optically clear nuclei with nuclear grooves and nuclear inclusions (Cibas and Ali, 2017). Together with examples of lung and oral mucosa shown in Fig. 2, cytological morphology from different organs provides diagnostic information for cancer screening and guides patient management at a minimal cost (Kontzoglou et al., 2005).

In cytology screening, scrutinizing individual cells or cell clusters under the microscope (or in gigapixel images) in search of malignancy can be quite time-consuming and tedious for cytologists (Ivanovic, 2014). Researchers have tried to develop accurate and efficient cancer screening systems to handle the ever-increasing caseload. The first successful trial could date back to the 1950s when an automatic system was developed for cervical screening of cytological smears (Tolles, 1955). Afterwards, many cervical screening systems were launched with great success (Koss et al., 1994; Wilbur et al., 2009). These automated screening systems have shown promise in improving efficiency without compromising accuracy.

Artificial intelligence (AI) has achieved remarkable progress in the field of medicine over the recent years. Machine learning (ML), a subfield of AI, focuses on learning algorithms to represent the underlying patterns in training data. With the rapid development of ML techniques, medical image interpretation and computer assisted-diagnosis have made significant advances in pathology (Lin et al., 2022a; Chen et al., 2016a), radiology (Lassau et al., 2021; Çallı et al., 2021), ophthalmology (Wang et al., 2020), etc. In computational cytology, previous studies have demonstrated the feasibility of various ML approaches in cytology image analysis, such as support vector machine (SVM), fuzzy c-means (FCM) (Chankong et al., 2014), k-means (Isa, 2005), and fuzzy clustering (Plissiti et al., 2009). However, there remain challenges in ML-based approaches, such as developing accurate and efficient cytology image analysis models, and establishing human–machine collaborative cytological screening systems.

Deep learning (DL), as a branch of the ML family, was developed with multilayer neural networks for leveraging feature representations of input data. The major advantage of DL algorithms is they can reduce the reliance on task-related features designed from expert knowledge and increase the feature representation capability by end-to-end learning. DL solutions in computational cytology are typically designed for accurate and efficient cancer screening, which have been extensively investigated in versatile cancer types, such as cervix (Rahaman et al., 2020), breast (Garud et al., 2017), bladder (Dov et al., 2021), and lung (Teramoto et al., 2017). These solutions typically build DL frameworks to learn the inferable function that maps cytological inputs into desired predictions. Most existing studies in DL-based cytology image analysis focus on network designs and knowledge integration for performance improvements, such as shape prior (Song et al., 2016), attention mechanism (Zhou et al., 2019b), and contour constraints (Song et al., 2018).

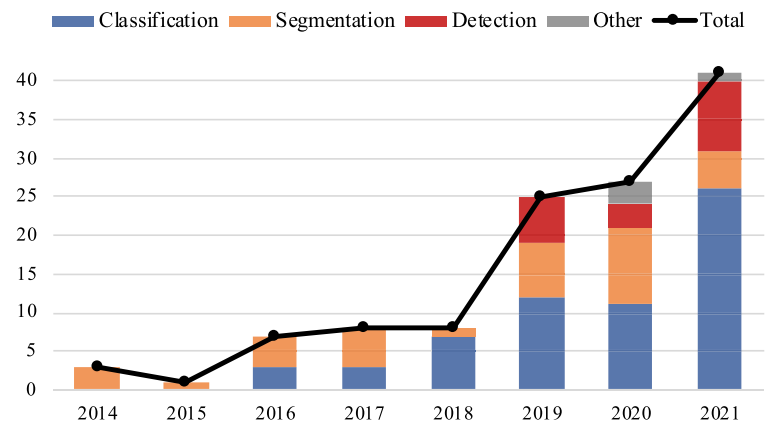

Fig. 3. Number of publications in DL-based computational cytology of classification, detection, segmentation and other tasks.

The advancement of DL has greatly accelerated the development of computational cytology. By searching literature in datasets (Google Scholar, PubMed, and arXiv) using search terms (cytology, cancer screening, Pap smear, deep learning, etc.), the number of related publications is illustrated in Fig. 3. It shows a 10-fold increase in DL-based computational cytology research from 2014 to 2021 with a booming trend in recent two years. These publications provide various DL approaches for cytological screening, such as classifying between normal and abnormal cells (Zhang et al., 2017), locating and identifying cells in cytological smears (Pirovano et al., 2021), and segmenting different cellular components (Zhou et al., 2020).

There exist several surveys in the field of computational cytology (Landau and Pantanowitz, 2019; Mitra et al., 2021; Rahaman et al., 2020). However, these reviews are far from exhaustive in terms of the advanced algorithms, publicly available datasets, challenges and opportunities in this field. Besides, most of the DL-based cytology surveys focused on the cervix, ignoring the progress in other types of cancer, such as lung, thyroid, and bladder (Rahaman et al., 2020). Mitra et al. (2021) offered a review of cytology image analysis from an application instead of methodology driven perspective. In this paper, we have surveyed over 120 publications since 2014, and systematically reviewed the progress of DL solutions in computational cytology. In addition, this survey covers cytology specimens from different parts of the anatomy, cancer types, and collection techniques.

There are six sections in this paper. Section 1 briefly introduces the background and objective of this review. Section 2 gives an overview of different learning paradigms in the context of computational cytology. Section 3 summarizes public cytology datasets and common evaluation metrics. Section 4 presents the progress and achievements in the DL-based cytology image analysis. Section 5 discusses existing challenges and potential research directions in computational cytology. Section 6 concludes this survey paper.

2. Deep learning methodology

This section presents the concepts, formulations, and general procedures of DL methods, we divide this section into: supervised learning (Section 2.1), weakly supervised learning (Section 2.2), unsupervised learning (Section 2.3), together with transfer learning (Section 2.4).

2.1. Supervised learning

Supervised learning aims to learn inferable mappings between input data and corresponding labels. In medical image analysis, the inputs are medical images, while the labels vary according to tasks, including image-level categories for classification, object-level localizations (e.g., boxes or points) for detection, and pixel-wise masks for segmentation. Formally, the input images $X = \{x_i\}_{i=1}^{N}$ together with corresponding labels $Y = \{y_i\}_{i=1}^{N}$ are used to train a predictive model by minimizing the objective function. Typical DL frameworks in supervised learning include multilayer perceptron (MLP) (Rosenblatt, 1961), convolutional neural network (CNN) (LeCun et al., 1998), recurrent neural network (RNN) (Zaremba et al., 2014), and Transformer (Dosovitskiy et al., 2020).

CNNs have been shown to be successful DL architectures in computer vision (Litjens et al., 2017). They are mainly constructed by three types of hidden layers: convolutional layers for extracting features, pooling layers for reducing the feature resolution, and fully connected layers for compiling the features extracted by previous layers and outputting prediction results. The backpropagation algorithm is utilized to update parameters of different layers during training (LeCun et al., 1998).

Commonly encountered CNN architectures, e.g., AlexNet (Mohammed et al., 2021), VGGNet (Albuquerque et al., 2021), ResNet (Miselis et al., 2019), have been employed and developed for medical applications. Currently, most cytology research activities focus on developing advanced algorithms based on these basic architectures in various DL tasks: classification (Lin et al., 2019b), detection (Xie et al., 2018), and segmentation (Matias et al., 2021), and others (Ma et al., 2021).

Classification. The classification model aims at predicting the category of the input image. It can be formulated as $\hat{y} = f(x, \theta)$, where x and $\hat{y}$ are the input image and predicted category, and θ represents learnable parameters of classification architecture. When training these models, cross-entropy loss L_{CE} is a commonly-used loss function to measure the discrepancy between predicted probability $\hat{y_{ic}}$ and true label y_{ic} by probability distribution (Bishop and Nasrabadi, 2006):

$$L_{CE} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{c=1}^{M} y_{ic}\log\left(\hat{y}_{ic}\right) \tag{1}$$

where, N is the number of images, and M is the number of categories. Then, the objective function together with an optimizer are used to update parameters θ of network by backpropagation (Rumelhart et al., 1986). Usually, the label space of the cytology image y_{ic} refers to normal/abnormal or sub-categories.

The standard cytology classification workflow is illustrated in Fig. 4. In cell-level classification, regions of interest (ROIs) are first extracted from collected whole slide images (WSIs). Then, these ROIs are cut into cell patches as input of the deep neural network (DNN). After that, a DNN-based feature extractor generates representative features and outputs prediction categories. For example, there are a variety of categories for cervical cell types, including low-grade squamous in the intraepithelial lesion (LSIL), high-grade squamous intraepithelial lesion

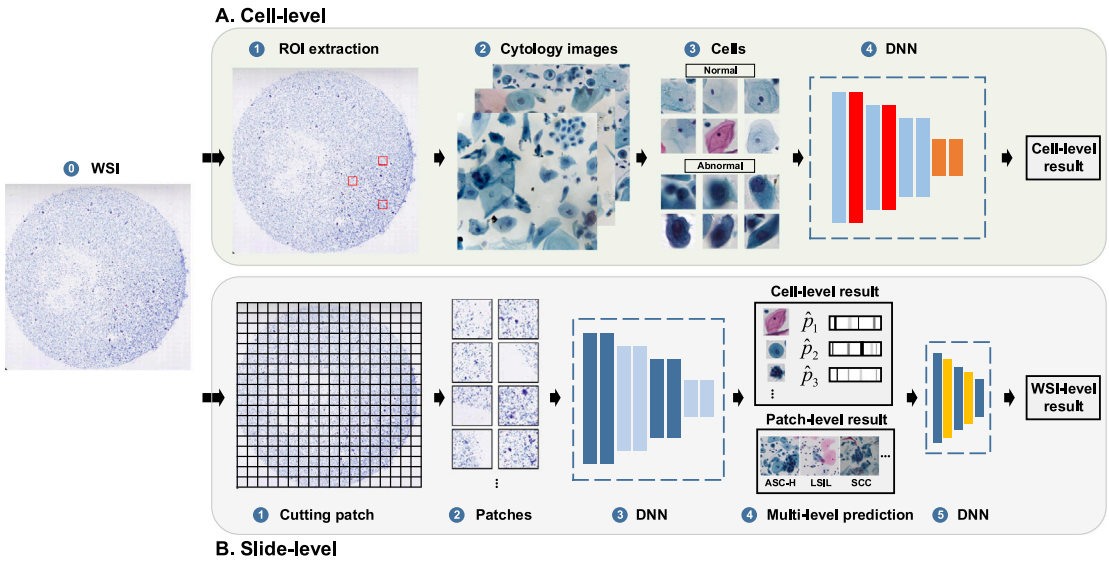

Fig. 4. Standard workflow of DL-based supervised learning for cytological classification. (A) Cell-level (top): Firstly, ROIs are extracted from WSIs and cut into cell patches. Then they are input into the DNN for extracting features and predicting the category of each cell patch. (B) Slide-level (bottom): Patches cut from WSIs are fed into a DNN to obtain multi-level predictions (e.g., cell-level and patch-level), then these predictions are aggregated to predict final slide-level results.

(HSIL), squamous cell carcinoma (SCC), atypical squamous cells cannot exclude HSIL (ASC-H), atypical squamous cells of undetermined significance (ASC-US), atypical glandular cells (AGC) (Nayar and Wilbur, 2015). For slide-level screening, as shown in Fig. 4(B), existing studies primarily divide this task into two stages: first, a DNN (e.g., CNN, RNN, and Transformer) provides multi-level predictions and information, such as detection of malignant or benign cells (cell-level), the category of patches (patch-level), and quantification of images (image-level). Subsequently, these predictions are aggregated by another network or diagnostic rules for assessing the malignancy risk grade of WSIs, hence providing patients with screening results (Wei et al., 2021; Lin et al., 2021).

Detection. The detection task is to locate objects from whole images (regression) and predict categories of these objects (classification), which can be divided into two categories in CNN-based algorithms: two-stage method and one-stage method. Two-stage detectors mainly follow the region proposal paradigm. Specifically, the first stage is to regress coarse prediction (box coordinates and predicted scores) by region proposal. The second stage aims to output final predictions by bounding box regression and classification (Girshick et al., 2014; Ren et al., 2015). Instead, one-stage object detectors tend to escape from the region proposal strategy toward directly predicting the category and location of the objects via end-to-end architectures (Liu et al., 2016; Redmon et al., 2016).

Training object detection models requires classification loss and regression loss. As a classification loss, focal loss L_{focal} is proposed in RetinaNet to address the issue of imbalance samples in cross-entropy loss L_{CE} (Lin et al., 2017b):

$$L_{focal} = \begin{cases} -\alpha(1-p)^{\gamma}\log(p), & \text{if} \quad y=1 \\ -(1-\alpha)p^{\gamma}\log(1-p), & \text{if} \quad y=0 \end{cases} \tag{2}$$

where p is the model's predicted probability with label y, and γ and α are tunable parameters. For bounding box regression, mean absolute error and mean square error are defined to calculate L_1 and L_2 distance, respectively. To overcome the sensitivity to outliers in L_1 loss and L_2 loss, a more robust loss, smooth L_1 loss is adopted to train regressors (Girshick, 2015):

$$\text{smooth}_{L_1}(x) = \begin{cases} 0.5x^2 & \text{if } |x|<1 \\ |x|-0.5 & \text{otherwise} \end{cases} \tag{3}$$

where x is defined as the difference between true and predicted bounding box coordinates. To avoid directly using independent points, the object detection evaluation metric, intersection over union (IoU), is designed as a regression loss (L_{IoU}) to calculate the IoU score between true and predicted bounding box.

$$L_{IoU} = -\ln\frac{\text{Intersection}\left(box_{gt}, box_p\right)}{\text{Union}\left(box_{gt}, box_p\right)} \tag{4}$$

where box_{gt} and box_p represent ground truth box and predicted box, respectively.

Segmentation. The fundamental and essential task in medical domain is to make the pixel-wise prediction which represents the morphology of biomedical structures, such as nuclei (Zhou et al., 2019c; Chen et al., 2016b, 2017), gland (Graham et al., 2019; Sirinukunwattana et al., 2017), and organ (Rahaman et al., 2020). Semantic segmentation predicts the category of each pixel to obtain masks of objects, which can be regarded as a pixel-wise classification task. Fully convolutional network (FCN) is one of the most successful segmentation architectures, which replaces the fully connected layers in traditional CNN with convolutional layers for outputting segmentation map (Long et al., 2015). Then, Falk et al. (2019) proposed U-Net for biomedical image segmentation by the U shape architecture and skip connection. In order to distinguish each segmented object, instance segmentation is to segment pixels into categories and assign them corresponding instance IDs, which is the task of simultaneously performing semantic segmentation and object detection. Most popular instance segmentation structures follow the top-down pipeline (Tian et al., 2019; He et al., 2017). For example, in the structure Mask R-CNN (Fig. 5), region proposal network (RPN) generates ROIs with predicted coordinates and scores in the first stage (Ren et al., 2015). Then, mask head is responsible for predicting the mask of each instance (He et al., 2017).

As the pixel-wise classification supervision, classification loss L_{CE} is applied to train segmentation networks. Another widely-used loss function, dice loss L_{dice} is designed to measure the similarity between predicted Y_p and ground truth Y_{gt} by calculating the dice similarity coefficient, defined as (Milletari et al., 2016):

$$L_{dice} = 1 - \frac{2|Y_p \cap Y_{gt}|}{|Y_p| + |Y_{gt}|} \tag{5}$$

2.2. Weakly supervised learning

Weakly supervised learning is appealing in medical image analysis scenarios where labels are not fully available, typically including

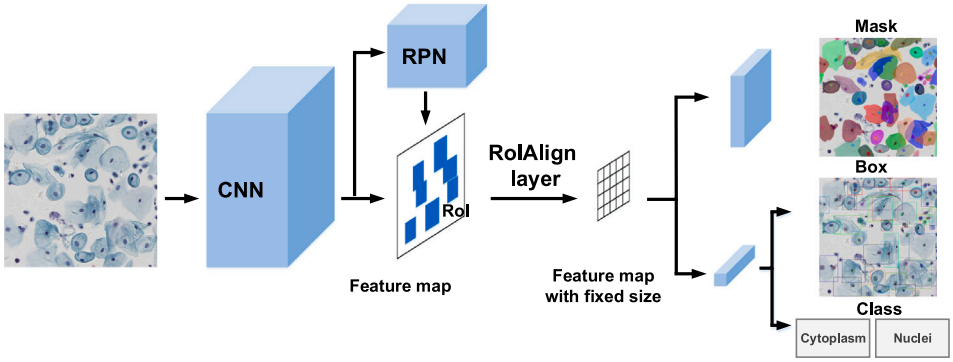

Fig. 5. Instance segmentation in cytology images by Mask R-CNN, which can predict the segmentation, detection and classification results in parallel.

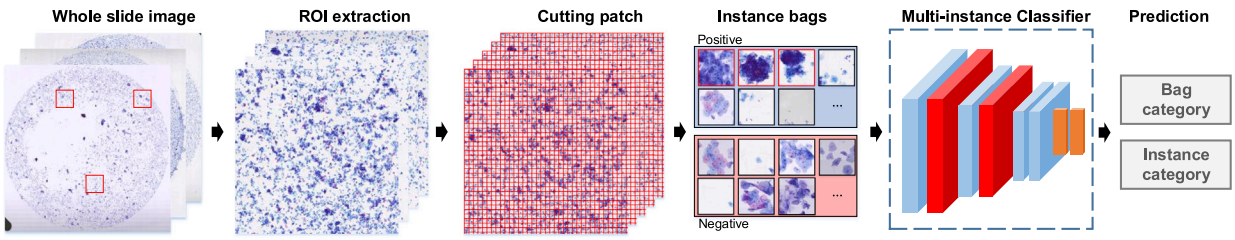

Fig. 6. Overview of the multiple instance learning paradigm. Original WSIs are extracted as ROIs and cut into patches, which are then formed into instance bags with bag-level labels (positive or negative). For learning algorithms, these bags are used to learn a classifier that can predict the labels of bags and instances in the testing data.

three types of weak supervision: incomplete supervision, inaccurate supervision, and inexact supervision (Zhou, 2018).

As a successful learning paradigm of incomplete supervision, semi-supervised learning is to leverage both labeled and unlabeled data by extracting hidden information from unlabeled sets to enhance the feature representation of the labeled set. For inaccurate supervision, methods for noisy label problem aim to identify the potentially mislabeled samples and make corrections, improving supervision reliability.

Multi-instance learning (MIL) is developed as an effective inexact supervision mechanism, which aims at utilizing coarse-level annotations (e.g., image-level) for learning fine-level (pixel-level and patch-level) tasks (Maron and Lozano-Pérez, 1998). The standard workflow of MIL is illustrated in Fig. 6. Firstly, a series of ROIs are extracted from whole images with patch-level annotations. Then, these ROIs are cut into patches and formed into bags. Finally, a multi-instance classifier is established by learning for multi-instance bags, which is used to predict unknown bags or instances. Specifically, given a training dataset $\{(X_i, Y_i)_{i=1}^N\}$, where $X_i = \{x_{i1}, x_{i2}, \ldots, x_{i,m}\}$ are instance bags, $x_{i,m}$ is the *m-th* instance of the *i-th* bag. $Y_i \in \{-1, +1\}$ is its corresponding label of the *i-th* bag, +1 represents positive bag with at least one positive instance in the bag, and −1 represents negative bag without positive instance. Subsequently, bags X_i together with labels are used to learn a classifier that can predict the labels of bags and instances.

2.3. Unsupervised learning

Unsupervised learning attempts to learn useful and underlying representations from unlabeled data for downstream tasks. For example, image augmentation techniques have been applied to increase the amount and variety of the original data, thereby improving the model performance (Chen et al., 2021b). Unsupervised image transformation can be applied to virtual staining and stain normalization (de Haan et al., 2021). The auto-encoder (AE) is a typical structure in unsupervised learning, which can be formulated as: $\mathrm{P}(x_i) \rightarrow z \rightarrow \mathrm{P}(x_i')$, where P denotes encoder and decoder. AE is trained to encode the input image x_i to obtain latent representation z. Then, the decoder generates reconstructed image x_i' with the supervision of the raw input x_i.

Two major unsupervised models have gained popularity: variational auto-encoder (VAE) (Larsen et al., 2016; Kingma and Welling, 2013) and generative adversarial network (GAN) (Goodfellow et al., 2014). As shown in Fig. 7(A), VAE improved AE by constraining latent variables to be normally distributed, then sampling a latent vector Z into a decoder for the final output. GAN is a type of generative model, which can potentially mitigate the effect of limited data by synthesizing high-quality fake images, as shown in Fig. 7(B). GAN is formed by a generator-discriminator architecture, in which the generator aims to generate realistic images while the discriminator competes with the generator to differentiate real images and generated images. Therefore, the generator-discriminator architecture is optimized via the adversarial training (Goodfellow et al., 2014):

$$\begin{aligned}\min_G \max_D V(D,G) = &\mathbb{E}_{x\sim p_{\text{data}}(x)}[\log D(x)] \\ &+\mathbb{E}_{z\sim p_z(z)}[\log(1 - D(G(z)))]\end{aligned} \tag{6}$$

where $G(\cdot)$ denotes the generator, and $D(\cdot)$ denotes the discriminator. $\mathbb{E}(\cdot)$ is the expectation value of distribution. $p_{\text{data}}(x)$ and $p_z(z)$ are the distribution of the real sample and noise, respectively.

2.4. Transfer learning

Transfer learning is a subfield of DL that focuses on transferring knowledge between cross domains to further improve the performance of task-specific models. When training DL models in the limited data scenario, the common practice is to pre-train a CNN on a large-scale dataset such as ImageNet (Krizhevsky et al., 2012) and then fine-tune the model using target data (Tan et al., 2018; Tajbakhsh et al., 2016). Fine-tuning is an appealing transfer learning solution in medical image analysis scenarios, which can strengthen the generalization capability and reduce the overfitting issue by transferring knowledge from natural image datasets to domain-specific tasks (Sornapudi et al., 2019; Yosinski et al., 2014). Formally, the goal of fine-tuning is learning a task $\mathcal{T}^t$ by a small dataset T. It exploits a large-scale and task-similar $\mathcal{T}^t$ dataset $S = \{s_i\}_{i=1}^M (M \gg N)$ to pre-train a network $f(\sim;\theta)$ first. Then, a small-scale target dataset $T = \{t_i\}_{i=1}^N$ is used to train the several last layers of the pre-trained model for obtaining the target model $f_t(\sim;\theta')$.

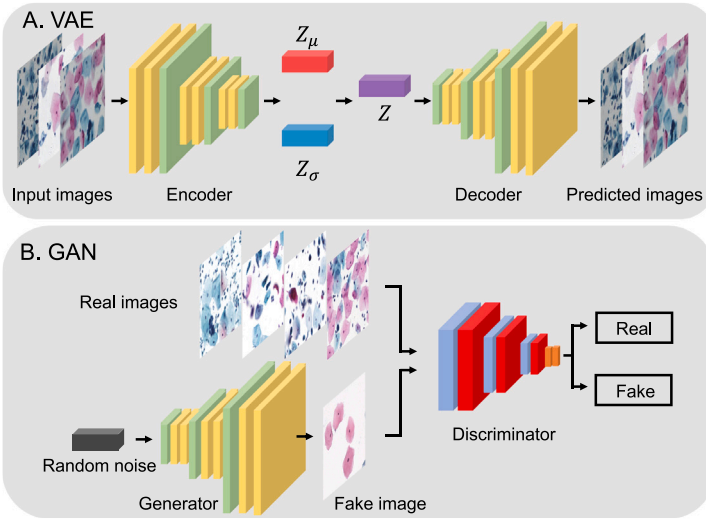

Fig. 7. AE-based unsupervised learning models. (A) VAE, an encoder–decoder architecture, Z_μ is mean vector, Z_σ is standard deviation vector, and Z is sampled latent vector. (B) GAN, a generator-discriminator architecture.

Under the premise that different datasets for learning similar tasks have similar low-level feature representations, fine-tuning is regarded as a common and effective training strategy in various DL image analysis applications as well as cytology image analysis (Zhang et al., 2017).

Domain adaptation (DA) is another transfer learning approach that transfers knowledge by narrowing the distribution gap between cross domain. DA can be typically depicted as: considering two domains, namely source domain (S) and target domain (T), with different data distributions ($p(S) \neq p(T)$). The distributions can be aligned through image-level and feature-level alignment, and transferrable knowledge can be learnt by target model for performance improvements (Oza et al., 2021; Guan and Liu, 2021).

3. Datasets and metrics

DL models rely on large amounts of labeled data, and we summarize publicly available datasets as well as representative private datasets. As illustrated in Table 1, the majority of public datasets are from the cervix with a relatively small number of other organs, like breast, oral. These publicly available cytology datasets can be exploited to develop DL algorithms for various tasks, including classification, detection, and segmentation.

Herlev (Jantzen et al., 2005). The database consists of 917 Papanicolaou (Pap) smear cervical images in 7 classes (3 normal and 4 abnormal cell classes), collected from the Herlev University Hospital. Herlev is the earliest established public cytology dataset for developing DL-based coarse and fine-grained classification models for cervical cancer screening (Zhang et al., 2017).

ISBI 2014 (Lu et al., 2015). Another widely used cervical dataset comes from the ISBI challenge. Unlike the Herlev dataset, the dataset focuses on the segmentation task with pixel-wise annotations. It consists of 16 non-overlapping fields of view images (40× magnification) with 645 cells. Each image in ISBI 2014 contains 20 to 60 Pap-stained cervix cells with varying degrees of overlapping.

ISBI 2015 (Lu et al., 2016). ISBI 2015 extends ISBI 2014 dataset to 945 images by synthesis. Cervical cytology images in ISBI 2015 have a varying number of cells and different degrees of cell overlapping (512 × 512 pixels), the dataset consists of 45 training images and 900 testing images (from 12 images).

Sipakmed (Plissiti et al., 2018). The dataset contains 4049 images with isolated cells, which were cropped 966 cluster cell images. It contains 5 types of cervical cells, including superficial-intermediate, parabasal, koilocytotic, dyskeratotic, and metaplastic cells.

CERVIX93 (Phoulady and Mouton, 2018). This is the first dataset established for nuclei detection tasks in cytology. It consists of 93 stacks of images at 40× magnification. Each stack has 10 to 20 images acquired in an equally spaced field of views from the top to the bottom of the slide. In this dataset, 2705 nuclei are annotated by bounding boxes with three Pap test grades: negative, LSIL, or HSIL.

FNAC (Saikia et al., 2019). This is the only surveyed public dataset for breast cytology image analysis. These breast images are collected from 20 patients, comprising 212 FNAC images in classes of benign (99) and malignant (113).

BHS (Araujo et al., 2019). It collects 194 Pap-smear cervical slides from the Brazilian Health System (BHS). There are 108 images with at least one abnormal cell and 86 images with only normal cells. In sum, it has 5 types of abnormalities: carcinoma, HSIL, LSIL, ASC-US, and ASC-H.

AgNOR (Amorim et al., 2020). The dataset is composed of 2540 images with 1200 × 1600 pixels each. Different from other datasets, it contains cervical cells stained with the silver technique, known as argyrophilic nucleolar organizer regions (AgNOR). For developing segmentation approaches, experts annotate objects by the LabelMe tool (Russell et al., 2008), including nuclei, clusters, and satellites.

LBC (Hussain et al., 2020a). Recently, liquid-based cytology (LBC) was developed to provide more uniform sample fixation techniques and cytology images with a cleaner background than traditional Pap smear tests. It consists of 963 images with four classes: NILM, LSIL, HSIL, and SCC.

Oral (Matias et al., 2021). Totally, 1934 oral images of 1200 × 1600 pixels are acquired from two Pap-stained slides of cancer diagnosed oral brush samples. With different types of annotation (class, box, mask), the dataset can be applied to develop classification, detection, and segmentation tasks.

CRIC (Rezende et al., 2021). The collection CRIC has 400 images of pap smears with 11,534 classified cells. Based on the Bethesda system (Nayar and Wilbur, 2015), CRIC dataset covers conventional cytology cervical cells with six types: NILM (6779), ASC-US (606), LSIL (1360), ASC-H (925), HSIL (1703), and SCC (161).

CDetector (Liang et al., 2021b). The dataset consists of 7410 cervical images cropped from the WSIs. According to the Bethesda system (TBS), 48,587 bounding boxes are annotated by experienced pathologists which belong to 11 categories: ASC-US, ASC-H, HSIL, LSIL, SCC, AGC, trichomonas, candida, flora, herps, and actinomyces. CDetector is the largest publicly available dataset for object detection in cytology until now.

Ascites (Su et al., 2020). The dataset was collected from Peking University for gastric cancer screening. It consists of 176 H&E stained

Table 1
Summary of publicly available and representative private databases in computational cytology.

Dataset/Year	Task	Organ	Stain	Size *(pixel)*	Description	Link
Herlev 2005 (Jantzen et al., 2005)	Classification	Cervix	Pap	Variable	917 cells	http://mde-lab.aegean.gr/downloads
ISBI 2014 (Lu et al., 2015)	Segmentation	Cervix	Pap	512 × 512	16 images (645 cells)	https://github.com/luzhi/cellsegmentation_TIP2015
ISBI 2015 (Lu et al., 2016)	Segmentation	Cervix	Pap	512 × 512	945 images synthesized by ISBI 2014	http://goo.gl/KcpLrQ
Sipakmed 2018 (Plissiti et al., 2018)	Classification	Cervix	Pap	2048 × 1536	966 images (4049 annotated cells)	https://www.cs.uoi.gr/~marina/sipakmed.html
CERVIX93 2018 (Phoulady and Mouton, 2018)	Classification detection	Cervix	Pap	1280 × 960	93 stacks of images (2705 nuclei)	https://github.com/parham-ap/cytology_dataset
FNAC 2019 (Saikia et al., 2019)	Classification	Breast	Pap	2048 × 1536	212 images in two classes: benign (99) and malignant (113)	https://1drv.ms/u/s!Al-T6d-_ENf6axsEbvhbEc2gUFs
BHS 2019 (Araujo et al., 2019)	Segmentation ranking	Cervix	Pap	1392 × 1040	194 images in classes of carcinoma, HSIL, LSIL, ASC-US, and ASC-H	https://sites.google.com/view/centercric
AgNOR 2020 (Amorim et al., 2020)	Segmentation	Cervix	AgNOR	1600 × 1200	2540 images (4515 nuclei)	https://arquivos.ufsc.br/d/373be2177a33426a9e6c/
LBC 2020 (Hussain et al., 2020a)	Classification	Cervix	Pap	2048 × 1536	963 LBC images in classes of NILM, LSIL, HSIL, and SCC	https://data.mendeley.com/datasets/zddtpgzv63/4
Oral 2021 (Matias et al., 2021)	Classification detection segmentation	Oral	Pap	1200 × 1600	1934 images (4287 annotations)	https://arquivos.ufsc.br/d/5035aec3c24f421a95d0/
Cric 2021 (Rezende et al., 2021)	Classification	Cervix	Pap	1376 × 1020	400 images (11,534 cells)	https://database.cric.com.br
CDetector 2021 (Liang et al., 2021b)	Detection	Cervix	Pap	224 × 224	7410 images (48,587 object instance bounding boxes) in 11 classes	https://github.com/kuku-sichuan/ComparisonDetector
Ascites 2020 (Su et al., 2020)	Classification detection	Stomach	H&E, Pap	1064 × 690	487 images for classification: malignant (18,558) and benign (6089); 176 images for detection (6573 bounding boxes)	https://pan.baidu.com/s/1r0cd0PVm5DiUmaNozMSxgg
RSDC 2021 (Ma et al., 2021)	Super resolution	Cervix	Pap	128 × 128 (HR) 64 × 164 (LR)	5 slides (25,000 patches)	https://www.kaggle.com/birkhoff007/rsdcdata
IRNet 2019 (Zhou et al., 2019b)	Segmentation	Cervix	Pap	1000 × 1000	413 images (4439 cytoplasm and 4789 nuclei)	Private dataset
DCCL 2020 (Zhang et al., 2019)	Detection	Cervix	Pap	1200 × 2000	1167 WSIs (14,432 patches, and 27,972 labeled lesion cells)	Private dataset
Dual 2021 (Lin et al., 2021)	Risk stratification	Cervix	Pap	Up to 50,000 × 50,000	19,303 WSIs in two classes: abnormal (202,557) and normal (272,933)	Private dataset
Hybrid 2021 (Zhu et al., 2021)	Classification detection segmentation	Cervix	Pap	6000 × 6000	24 categories and 2000 images in each category, 81,727 smears (1.7 million annotated targets) for detection model	Private dataset

and Pap stained images cropped from ascites cytopathology images at 40× magnification. A total of 6573 cells (benign and malignant) were annotated using bounding boxes.

RSDC (Ma et al., 2021). It is the only public cytology dataset for the refocusing and super-resolution task. The images in the dataset are collected from 5 LBC slides with the resolution 0.243 μm/pixel. The dataset contains 15,000 low-resolution images (64 × 64 pixels) and corresponding high-resolution images (128 × 128 pixels) generated from original slides.

Additionally, we summarize the evaluation metrics in terms of three canonical DL tasks in cytology image analysis: classification, detection, and segmentation. More details are available in Table 2.

Different metrics are used to evaluate the performance of the model from different perspectives. Among these summarized classification metrics, the $Accuracy$ is the most straightforward metric, yet it ignores the class imbalance problem. The confusion matrix can represent the prediction result for each category. To measure object detection tasks, IoU is a commonly-used metric, which can measure the degree of

Table 2
Summary of evaluation metrics in computational cytology.

Metric	Definition	Description	Application in cytology
Classification			
TP/TN/FP/FN	True Positive, True Negative, False Positive, False Negative.	A test result that correctly indicates the presence of a condition (TP), correctly indicates the absence of a condition (TF), wrongly indicates the presence of a particular condition (FP), wrongly indicates the absence of a particular condition (FN).	Classification of FNAC images; formulate other metrics (e.g., accuracy, precision, and recall) (Sanyal et al., 2018; Liang et al., 2021a).
Confusion matrix	A Matrix. Row: GT; Column: prediction.	The number of correct and incorrect predictions are summarized with count values and broken down by each class.	Classification of lung cancer sub-type; quantitative analysis of abnormalities; WSI-level risk stratification (Teramoto et al., 2017; Mohammed et al., 2021; Ke et al., 2021; Awan et al., 2021).
Accuracy (Acc)	$\frac{TP+TN}{FP+FN+TP+TN}$	Proportion of all positive and negative classes with correct predictions in all samples.	Cervical squamous lesions classification; FNAC image classification (Liu et al., 2020; Bal et al., 2021; Albuquerque et al., 2021).
Precision (P)	$\frac{TP}{FP+TP}$	The proportion of positive samples classified as positive examples by the classifier.	Cervical cells classification; cervical lesions classification; multi-cell classification in liquid-based cytology images (Sornapudi et al., 2019; Liu et al., 2020; Rahaman et al., 2021).
Recall (R)	$\frac{TP}{FN+TP}$	The proportion of the samples predicted to be positive cases in the total positive cases.	Pap smear image classification; classification of cervical cells (Mohammed et al., 2021; Rahaman et al., 2021).
Specificity (Spec)	$\frac{TN}{FP+TN}$	The proportion of samples that are correctly predicted as negative classes in all negative classes.	Cell classification; differential diagnosing of papillary thyroid carcinomas; detection of cervical intraepithelial neoplasia or invasive cancer (Zhang et al., 2017; Guan et al., 2019b; Bao et al., 2020).
Sensitivity (Sens)	$\frac{TP}{FN+TP}$	The proportion of the samples predicted to be positive cases in the total positive cases.	Distinguish large cell neuroendocrine; identify cells; high resolution image classification (Gonzalez et al., 2020; Li et al., 2021).
F1-score (F1)	$\frac{2\times Precision\times Recall}{Precision+Recall} = \frac{2\times TP}{FP+FN+2TP}$	Harmonic average of precision and recall, and it is defined as the final evaluation index in many classification tasks.	Classification of cervical cells; multi-cell classification in liquid-based cytology images; cell image ranking (Sornapudi et al., 2019; Araujo et al., 2019; Rahaman et al., 2021).
ROC curve	(FP rate, TP rate)	ROC is a graph showing the performance of a classification model at all classification thresholds.	Prediction of malignancy; cervical cancer screening; smear-level risk stratification (Elliott Range et al., 2020; Tan et al., 2021; Lin et al., 2021).
AUC	Area under the ROC curve	The closer the AUC is to 1, the better the classifier performance.	Cancer screening (cell-level detection, patch-level and case-level classification); quantitative analysis of abnormalities (Vaickus et al., 2019; Ke et al., 2021; Cao et al., 2021).
Detection			
IoU	$\frac{P\cap GT}{P\cup GT} = \frac{TP}{FP+FN+TP}$	P denotes predicted bounding box, and GT is ground truth box. The ratio of the intersection and union of the predicted bounding box and the ground truth bounding box.	Nuclei/Cell detection; automation-assisted cervical cancer reading (Kilic et al., 2019; Xiang et al., 2020; Liang et al., 2021a).
AP	Average precision	The mean value of precision on precision–recall curve.	Detection of abnormal cervical cells (Cao et al., 2021).
mAP	mean AP	Average of AP in all categories.	Cell/Clump detection; pulmonary hemosiderophages quantification (Marzahl et al., 2020; Chai et al., 2022b; Liang et al., 2021b).
Frames Per Second (FPS)	$\frac{N}{t}$	N denotes the amount of test frame (image), and t is the test time (second).	Cervical cell/clumps detection (Liang et al., 2021b)
Segmentation			
Pixel precision (P_p)	$\frac{TP_p}{TP_p+FP_p}$	The p means this is a pixel-level metric. Proportion of correctly segmented pixels to all segmented pixels.	Cytological examination (overlapping cell segmentation) (Tareef et al., 2017).
Pixel recall (R_p)	$\frac{TP_p}{TP_p+FN_p}$	Proportion of correctly segmented pixels to all pixels in the ground truth.	Cytological examination (overlapping cell segmentation) (Tareef et al., 2017).
Pixel accuracy (Acc_p)	$\frac{TP_p+TN_p}{FP_p+FN_p+TP_p+TN_p}$	Pixel level accuracy.	Segmentation of cytoplasm and nuclei (Song et al., 2015, 2014; Ke et al., 2021).

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overlap between the predicted and the ground-truth box. Under different IoU thresholds, average precision (AP) is utilized to evaluate the performance of detectors. For example, AP_{50} means the value of the metric AP when IoU is set to 0.5. Among evaluation metrics for segmentation tasks, $Pixel\ accuracy$ measures the predicted result of each pixel, and $Dice$ calculates the similarity coefficient between predicted and ground truth masks.

4. Deep learning in cytology application

In this section, we survey and summarize literatures on various DL models applied to computational cytology. Firstly, we introduce preprocessing techniques in cytology image analysis, followed by typical tasks: classification, detection, segmentation, and others. More details of these surveyed literatures are as follows.

Table 2 (*continued*).

Metric	Definition	Description	Application in cytology
Hausdorff distance	$\max(\sup_{x\in X} d(x,Y), \sup_{y\in Y} d(X,y))$	X and Y are two sets, *sup* represents the supremum. It measures the similarity between two point sets.	Cell nuclei segmentation (Kowal et al., 2020).
Dice coefficient (Dice)	$\frac{2\times TP}{FP+FN+2TP}$	Dice coefficient is a statistical tool which measures the similarity between two sets of data. It can be used for comparing algorithm output against reference masks.	Semantic instance segmentation of touching and overlapping objects; cytoplasm segmentation; instance segmentation (Böhm et al., 2019; Wan et al., 2019; Walter et al., 2021).
Zijdenbos similarity index (ZSI)	$2\frac{\lvert R_{GT}\cap R_{S_{eg}}\rvert}{\lvert R_{GT}\rvert+\lvert R_{S_{eg}}\rvert}$	R_{GT} and $R_{S_{eg}}$ denote the ground truth and segmented regions, respectively. ZSI computes the ratio of aggregated union between cardinality predicted segmentation output and manual segmentation output.	Overlapping cell segmentation; cervical nuclei segmentation (Tareef et al., 2017; Hussain et al., 2020b).
Average Jaccard Index (AJI)	$\frac{\sum_{i=1}^{N}\lvert G_i\cap P_M^i\rvert}{\sum_{i=1}^{N}\lvert G_i\cup P_M^i\rvert+\sum_{F\in U}\lvert P_F\rvert}$	G_i is the ith object from the ground truth with N objects. P_M^i means the Mth connected component in prediction which has the largest Jaccard Index with G_i. AJI measures the ratio of the aggregated intersection and aggregated union for all the predictions and ground truths in the image.	Cell segmentation (Zhou et al., 2019b).

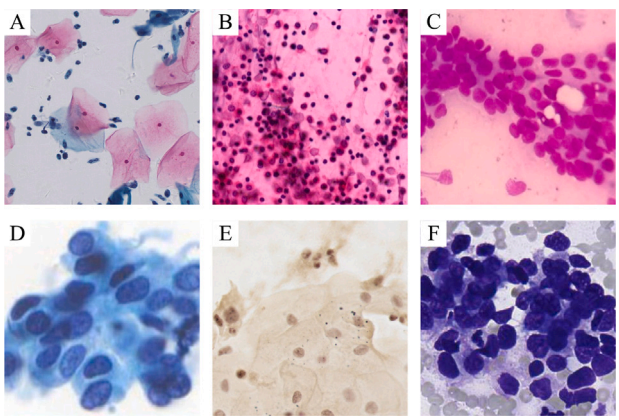

Fig. 8. Staining techniques: (A) Pap (Zhou et al., 2019b), (B) H&E (Su et al., 2020), (C) Giemsa (Bal et al., 2021), (D) Feulgen (Xiang et al., 2020), (E) AgNOR (Amorim et al., 2020), (F) Diff-Quik (Gonzalez et al., 2020).

4.1. Preprocessing

Staining techniques. In cytology, staining techniques are introduced to enhance representative features of cells (e.g., texture, structure, and other biochemical properties) for visually presenting cellular structures.

Fig. 8 shows various staining techniques in cytology. (1) Pap. As an extensive staining protocol, there are four steps: fixation, nuclear staining, cytosol staining, and transparency. Cells stained by Pap have clearly-structured nuclei and transparent cytoplasms. According to surveyed literatures, Pap is the most commonly encountered staining method in cytology, especially for cervical cytology images. (2) Hematoxylin and Eosin (H&E). Hematoxylin stains cell nuclei purplish-blue, and eosin stains the extracellular matrix and cytoplasm pink. In clinical practice, H&E is mainly used to stain cells in cytology slides and tissues in histology slides. (3) Giemsa. It is particularly effective for staining the cytoplasm, and hence is mainly employed in blood and bone marrow cytology. Some special situations require other staining techniques. For example, Amorim et al. (2020) stained cervical cells by AgNOR to show cell proliferation, differentiation, and malignant transformation. Xiang et al. (2020) stained cervical specimens of LBC by Feulgen, and Marzahl et al. (2020) stained pulmonary hemosiderophages cytology using Perlss' Prussian Blue and Turnbull's Blue.

Stain Normalization. Typically, there always exist diverse stain variations between different cytology images due to batch effects under different experimental conditions, including staining durations, imaging environments, and imaging protocols. These stain variations make it difficult to build robust and generalized DL-based cytology models (Tellez et al., 2019; Dawood et al., 2021). In order to address this issue, stain normalization can be a crucial preprocessing step in DL-based studies, especially when analyzing stained images, such as cytology and histopathology images (Khan et al., 2014). Traditional stain normalization techniques include whiting for removing redundant information from input data, and linear normalization for scaling gray values of input data, etc. After that, DL-based solutions have been designed to address this issue, such as style transformation, and stain estimation (Teramoto et al., 2021; Zhou et al., 2019a).

Data augmentation. Insufficient data and limited diversities are influential factors that can hinder the development of robust and generalized DL models. Data augmentation strategies can increase the amount and diversity of training data, boosting the generalization of DL models. Conventional augmentation methods include geometric transformation (e.g., flipping, rotating, and scaling) and color transformation (e.g., noise, blurring, and contrast). Recently, techniques such as adversarial training have been adopted to synthesize large-scale datasets based on limited sets (Lin et al., 2022b). For example, Teramoto et al. (2020) proposed a GAN-based augmentation structure, called progressive growing of GANs (PGGAN). In this study, real lung cytological images together with generated images were used to train CNNs, leading to performance improvements in cytology image classification.

4.2. Classification

In cytology, most DL approaches are feasible and promising for image classification with distinguishable characteristics of cytology samples. In clinical workflows, cytologists can distinguish between benign and malignant cells based on their cytological features. Specifically, abnormalities are displayed in malignant cells, such as larger and irregular nuclei, an enlarged nuclear-cytoplasmic ratio, and the altered shape of the nucleolus. Within cytology classification tasks, DL-based models aim to extract the underlying and recognizable patterns of input images for identifying cell-level objects (e.g., nucleus and cell) or making slide-level predictions (e.g., cytopathology screening). Therefore, we further divide the cytological classification task into two categories: (1) cell-level classification, and (2) slide-level classification (see Table 3).

4.2.1. Cell-level classification

Cell-level classification has been one of the most successful tasks in DL-based cytology image analysis (Jantzen et al., 2005). While training DL networks for classifying cells, there is a need to cut scanned cytology slides into cell patches due to the giga-pixel size of the WSIs (Zhang et al., 2019). Then, these patches are fed into the DL network for cell-level classification.

Table 3
Overview of deep learning-based classification studies for computational cytology.

Reference	Application	Staining	Organ	Method	Dataset	Result
Cell-level classification						
Zhang et al. (2017)	Cell classification	Pap H&E	Cervix	Data preprocessing (patch extraction, data augmentation) + CNN + Transfer learning (fine-tune)	Herlev; HEMLBC	Herlev: Sens = 0.982, Spec = 0.983, Acc = 0.983; HEMLBC: Sens = 0.983, Spec = 0.990, Acc = 0.986.
Teramoto et al. (2017)	Classification of cancer types (adenocarcinoma, SCC, small cell carcinoma)	Pap	Lung	Data augmentation + CNN	Private dataset: 76 images in classes of adenocarcinoma (40), SCC (20), small cell carcinoma (16)	Adenocarcinoma: Acc = 0.89; SCC: Acc = 0.600; Small cell carcinoma: Acc = 0.703; Total: Acc = 0.711.
Dimauro et al. (2019)	Cell classification	Pap	Nasal	Three-block CNN	Private dataset: 3423 images (cell)	Sens = 0.97, Acc = 0.99.
Shanthi et al. (2019)	Malignancy detection and classification	Pap	Cervix	Three-layer CNN	Herlev	5-class: Acc = 0.941; 4-class: Acc = 0.962; 3-class: Acc = 0.948; 2-class: Acc = 0.957.
Teramoto et al. (2019)	Classification of benign and malignant cells	Pap	Lung	Data augmentation + VGG-16 + Grad-CAM	Private dataset: 621 images (patch) in classes of benign (306) and malignant (315)	Patch-level: Acc = 0.792, AUC = 0.872; Case-level: Acc = 0.870, AUC = 0.932.
Liu et al. (2020)	Classification of cervical squamous lesions	Pap	Cervix	VGG-16	Private dataset: abnormal cells (1736) and normal cells (1554)	Acc = 0.9807, P = 0.9791, Sens = 0.9801, F1 = 0.9809.
Lin et al. (2019b)	Fine-grained cell classification	Pap	Cervix	Fine-tune + CNN (AlexNet, GoogLeNet, ResNet, DenseNet)	Herlev	GoogLeNet: Acc = 0.945 (2-class), Acc = 0.713 (4-class), Acc = 0.645 (7-class).
Sornapudi et al. (2019)	Multi-cell classification in liquid-based cytology images	Pap	Cervix	Fine-tune + CNN (ResNet-50, VGG-19, DenseNet-121, Inception-v3)	Herlev; Private dataset: 25 images	ResNet-50: AUC = 0.95; VGG-19: AUC = 0.95; DenseNet-121: AUC = 0.94; Inception-v3: AUC = 0.88.
Hussain et al. (2020c)	Cervical cancer diagnostic prediction	Pap	Cervix	CNN (AlexNet, VGG-16, VGG-19, ResNet-50, ResNet-101, GoogLeNet)	Herlev	AlexNet: Acc = 0.8; VGG-16: Acc = 0.8337; VGG-19: Acc = 0.8455; ResNet-50: Acc = 0.8937; ResNet-101: Acc = 0.9450; GoogLeNet: Acc = 0.9567.
Mohammed et al. (2021)	Smear classification	Pap	Cervix	10 popular pre-trained CNN	Sipakmed	DenseNet-169: Acc = 0.990, P = 0.974, R = 0.974, F1 = 0.974.
Albuquerque et al. (2021)	Classification of cervical cancer risk	Pap	Cervix	9 popular CNN	Herlev	Acc = 0.756 (7-class), Acc = 0.813 (4-class).
Miselis et al. (2019)	Classification of FANC images	H&E	Breast	CNN (AlexNet, GoogLeNet, SqueezeNet, DenseNet, Inception-v3)	Private dataset: 737 images (ROIs from specimens) in classes of benign (275) and malignant (462)	AlexNet: AUC = 0.9730; GoogLeNet: AUC = 0.9455; SqueezeNet: AUC = 0.9152; DenseNet: AUC = 0.9244; Inception-v3: AUC = 0.9730.
Manna et al. (2021)	Classification of cervical cells	Pap	Cervix	Fuzzy rank + Pre-trained CNN (Inception-v3, Xception, DenseNet-169)	Sipakmed	Acc = 0.9855, Sens = 0.9852.
Rahaman et al. (2021)	Classification of cervical cells	Pap	Cervix	Hybrid deep feature fusion + CNN (VGG-16, VGG-19, XceptionNet, ResNet-50)	Sipakmed	Acc = 0.9985 (2-class), Acc = 0.9914 (3-class), Acc = 0.9914 (5-class).
Yu et al. (2021)	Classification of cervical cells	Pap	Cervix	AlexNet + GAN	Private dataset: 22,124 images (cell) in classes of abnormal (1202) and normal (20,922)	AUC = 0.984
Dey et al. (2019)	FNAC cytology image classification	H&E	Breast	Conditional GAN (synthesis) + CNN (ResNet-152, DenseNet-161, Inception-v3)	Private dataset: 150 images in classes of begin (75) and malignant (75)	ResNet-152: Acc = 0.7667; DenseNet-161: Acc = 0.8667; Inception-v3: Acc = 0.8000.
Teramoto et al. (2020)	Classification of cytological images	Pap	Lung	CNN + PGGAN	Private dataset: 511 images (patch) in classes of benign (244) and malignant (267)	Acc = 0.853, Sens = 0.854, Spec = 0.853.
Bakht et al. (2020)	Classification of FNAC images	Pap	Thyroid	CNN (VGG-19, AlexNet) + Transfer learning (Fine-tune)	Private dataset: 9209 images (cell) in 5 classes	VGG-19: Acc = 0.9305; AlexNet: Acc = 0.9288.
Kaneko et al. (2021)	Cell image recognition	Pap	Urine	EfficientNet	Private dataset: 4637 images (cell)	Acc = 0.95, Sens = 0.97, Spec = 0.95, AUC = 0.99.

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Table 3 (*continued*).

Reference	Application	Staining	Organ	Method	Dataset	Result
Żejmo et al. (2017)	Classification of cancer cytological specimen	H&E	Breast	CNN (AlexNet, GoogLeNet)	Private dataset: 550 images (ROIs) in classes of malignant (275) and benign (275)	AlexNet: Acc = 0.80; GoogLeNet: Acc = 0.83.
Garud et al. (2017)	Classification of FANC cells	H&E	Breast	GoogLeNet	Private dataset: benign (24) and malignant (13)	Acc = 0.8076.
Vaickus et al. (2019)	Automating the Paris system for cytopathology	Pap	Urine	VGG-19 + Morphometric model	Private dataset: negative (51), atypical (60), suspicious (52), positive (54)	Acc = 0.972, Spec = 0.976, Sens = 0.970.
Shi et al. (2021)	Classification of cervical cells	Pap	Cervix	Graph convolutional Network (GCN)	Sipakmed	Acc = 0.9837, Sens = 0.9980, Spec = 0.9960, F1 = 0.9980.
Saikia et al. (2019)	Classification of FNAC images	Pap	Breast	CNN (VGG-16, VGG-19, ResNet-50, GoogLeNet-v3)	FANC 2019	VGG-16: Acc = 0.8867; VGG-19: Acc = 0.8820; ResNet-50: Acc = 0.9056; GoogLeNet-v3: Acc = 0.9625.
Nojima et al. (2021)	Diagnose malignant potential of carcinoma cells	Pap	Urine	Visual geometry group CNN	Private dataset: 203 images	AUC = 0.9890, F1 = 0.9002.
Lilli et al. (2021)	Cell classification	Pap	Urine	VGG-16	Private dataset: 690 images in classes of urothelial normal (274) and abnormal cells (416)	Acc = 0.899.
Guan et al. (2019b)	Differential diagnosing of papillary thyroid carcinomas	H&E	Thyroid	VGG-16 and Inception-v3	Private dataset: 279 images (thyroid nodules)	VGG-16: 0.97 (image-level), 0.95 (patient-level); Inception-v3: 0.92 (image-level), 0.87 (patient-level).
Bal et al. (2021)	Classification of FNAC images	Giemsa H&E	Breast	CNN (13 layers, convolution and fully-connected layers)	Private dataset: 1020 Giemsa-stained images and 631 H&E-stained images	Giemsa: Acc = 0.9781, Spec = 0.982, F1 = 0.975; H&E: Acc = 0.9753, Spec = 0.987, F1 = 0.961.
Guan et al. (2019a)	Differential diagnosing of lymph node	H&E	Cervix	Inception-v3	Private dataset: 742 images in 4 classes	Acc = 0.8962.
Bhatt et al. (2021)	Cervical cancer detection	Pap	Cervix	EfficientNet + Grad-CAM	Herlev, Sipakmed	Acc = 0.9970, P = 0.9970, R = 0.9972, F1 = 0.9963, Kappa = 0.9931.
Noyan et al. (2020)	Cell identification	Giemsa	Skin	ResNet-50	Private dataset: 2260 images (Tzanck smear)	Acc = 0.943, Sens = 0.837, Spec = 0.973, AUC = 0.974.
Bao et al. (2020)	Detection of cervical intraepithelial neoplasia or invasive cancer	Pap	Cervix	VGG-16	Private dataset: 188,542 mages	CIN 2: Acc = 0.926; CIN 3+: Acc = 0.961.
Sanyal et al. (2018)	Classification of FNAC images	Giemsa Pap	Thyroid	CNN	Private dataset: 370 images in classes of non-PTCA (184) and PTCA (186)	Sens = 0.9048, Spec = 0.8333, Acc = 0.8506.
Wu et al. (2018)	Classification of cancer types	H&E	Cervix	AlexNet	Private dataset: 79 specimens in 3 classes	Acc = 0.9333.
Li et al. (2022)	Cervical cell classification	Pap	Cervix	ResNet-50 + Attention mechanism + LSTM	Sipakmed	Sensitivity = 0.999, specificity = 0.998, F1 = 0.9989.
Slide-level classification						
Elliott Range et al. (2020)	Classification (prediction of malignancy)	Pap	Thyroid	AlexNet	Private dataset: 908 WSIs	Sens = 0.92, Spec = 0.905, AUC = 0.932.
Dov et al. (2019)	Classification (prediction of malignancy)	Pap	Thyroid	VGG-11 + Multiple instance learning	Private dataset: 908 WSIs	AUC = 0.932, AP = 0.872.

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The most straightforward method is directly feeding cell patches into a multi-layer CNN to extract features, then cross output layers to obtain the predicted category. For example, Teramoto et al. (2017) designed a deep CNN with only three convolutional layers, three pooling layers, and two fully connected layers. In the experiments of lung cancer cytology classification (76 cases), about 70% of cell images were correctly classified, which is on par with the performance of cytologists in the experiments (Nizzoli et al., 2011). This study demonstrates the capability of simply-designed CNNs to construct cytological classification models, and the potential of DL-based models in cytological image analysis. However, such CNNs constructed by a small number of network layers suffer from limited feature representation capabilities

Table 3 (*continued*).

Reference	Application	Staining	Organ	Method	Dataset	Result
Tan et al. (2021)	Cervical cancer screening	Pap	Cervix	Faster R-CNN	Private dataset: 408,030 images	Sens = 0.994, Spec = 0.348, AUC = 0.67.
Ke et al. (2021)	Quantitative analysis of abnormalities	Pap	Cervix	U-Net, ResNet-50	Private dataset: 130 WSIs	Segmentation: Pixel Acc = 0.974, IoU = 0.913; Classification: Acc = 0.945.
Cao et al. (2021)	Cancer screening (cell detection, patch-level and case-level classification)	Pap	Cervix	Multi-scale region-based CNN + Attention mechanism	Private dataset: 7030 images	Classification: AUC = 0.990 (patch-level), AUC = 0.934 (case-level); Detection: AP = 0.75.
Li et al. (2021)	High resolution image classification	Pap	Cervix	Mixed supervision learning (image-level + pixel-level)	Private dataset: 862 images from 2 data centers	Center A: Sens = 1, Spec = 0.86; Center B: Sens = 1, Spec = 0.87.
Dov et al. (2021)	Classification (prediction of malignancy)	Pap	Thyroid	MIL+ NoisyAND + Attention mechanism + Maximum likelihood estimation	Private dataset: 142 WSIs with 4494 instances	AUC = 0.870 ± 0.017, AP = 0.743 ± 0.037.
Gonzalez et al. (2020)	Distinguish large cell neuroendocrine	Pap, H&E, Diff-Quik	Lung	CNN	Private dataset: 40 images in high-grade neuroendocrine carcinoma (17 small, 13 large, 10 mixed cells)	H&E: Acc = 0.900; Pap: Acc = 0.875; Diff-Quik: Acc = 0.889.
Zhu et al. (2021)	Rapid TBS classification of cervical liquid-based thin-layer cell smears	Pap	Cervix	Model assembly: Xception, YOLOv3, U-Net	Private dataset: 81,727 images	Speed = 180 s/slide, Sens = 0.9474.
Wei et al. (2021)	Lesion detection, WSI-level classification	Pap	Cervix	YOLOv3 + Transformer	Private dataset: 2019 slides from four scanning devices	AUC = 0.872.
Cheng et al. (2021)	WSI-level cervical cancer screening	Pap	Cervix	CNN (ResNet-50) + RNN	Private dataset: 3545 slides with 79,911 annotations	Spec = 0.935, Sens = 0.951, Speed = 1.5 min/slide.
Awan et al. (2021)	Cell-level classification, WSI-level risk stratification	Pap	Urine	RetinaNet + Counting of atypical and malignant cells	Private dataset: normal (243), inflammatory (13), CA (76), ASM (38), TCC (28)	Cell-level classification: AUC = 0.99; Risk stratification: AUC = 0.83.
Lin et al. (2021)	Smear-level risk stratification	Pap	Cervix	CNN with dual-path encode + Synergistic grouping loss	Private dataset: 19,303 WSIs from 4 centers in 6 classes	Sens = 0.907, Spec = 0.80, AUC = 0.925.

in complex classification tasks (e.g., fine-grained classification), easily leading to some issues, such as the lack of generalizability, overfitting of specific domains, and network non-convergence (Mhaskar et al., 2017). To learn better representations of cytology images, researchers then developed deeper networks with more layers. For instance, Zhang et al. (2017) proposed a CNN with eleven hidden layers for cervical cell classification. This study used transfer learning to learn a more generalized model for classifying seven types of cervical cells. It was first pre-trained on ImageNet (Krizhevsky et al., 2012), then fine-tuned using Herlev Pap smear dataset (Jantzen et al., 2005) and HEMLBC (H&E stained manual liquid-based cytology) dataset (Zhang et al., 2014), respectively. The results showed higher classification accuracies than CNNs with fewer layers or without pre-training.

In recent years, many advanced DL models have been proposed in the computer vision field, such as Inception (Szegedy et al., 2015), ResNet (He et al., 2016) and DenseNet (Huang et al., 2017). With the increase of network layers and complex structure designs (e.g., residual modules and Inception modules), these advanced CNNs have stronger visual feature representation capabilities, which is promising to improve the performance of cytology classification models. For example, Teramoto et al. (2019) employed VGG architecture with 16 layers to classify benign and malignant cells from lung cytology images. Besides, Noyan et al. (2020) proposed the TzanckNet based on ResNet-50 to identify cells in the cytology of erosive-vesiculobullous diseases. These studies showed that these advanced CNNs performed better in specific scenarios. They can be easily adopted to develop cytological image analysis algorithms, benefiting the technical and application development of computational cytology. Additionally, some studies implemented experiments to compare the performance of advanced CNN architectures in cytology image classification (Sornapudi et al., 2019; Hussain et al., 2020c; Mohammed et al., 2021; Albuquerque et al., 2021). From their results, popular architectures (e.g., ResNet, Inception, and DenseNet) achieved promising performance in cell-level classification tasks.

With the high performance achieved by advanced algorithms, building explainable and understandable AI systems necessitates further interpretability analysis, as this affects the reliability of decision-making in the clinic (Tjoa and Guan, 2020). Grad-CAM can identify decision-making related regions for interpretability analysis by visualizing the input gradient of the final convolutional layer (Selvaraju et al., 2017). While developing cytology classification approaches, Selvaraju et al. (2017) designed Grad-CAM to show region of interests (ROIs) in network decision-making using the gradient information of the last convolution layer of CNN. Similarly, Nojima et al. (2021) aimed to provide support for the diagnostic prediction of DL-based models by analyzing heatmaps generated via Grad-CAM and cytological properties (nuclear morphology, nuclear-cytoplasmic ratio, etc.).

Apart from binary classification (i.e., benign and malignant), multi-class scenarios are more common, yet challenging in cytology image analysis, since benign and malignant cells may contain several subcategories (Vaickus et al., 2019). For example, Herlev dataset contains 3 types of normal and 4 types of abnormal cervical cells (Jantzen et al., 2005). However, the classification decision boundaries of two subcategories of cells in the feature space are usually ambiguous, which brings challenges for CNNs to learn discriminative features. To develop a fine-grained classification model for classifying 7 subtypes of cervical cells, Lin et al. (2019b) introduced mask maps as the morphological appearance information to further enhance fine-grained features of cells, which could be beneficial to improve the classification accuracy.

In addition, several cytological studies focus on developing DL approaches in limited or imbalanced datasets. GAN-based models can be utilized to augment the original dataset for the performance improvement of the classification task. For example, Dey et al. (2019) synthesized 180 images by conditional GAN. Together with the original data, these images are utilized for training several popular models (ResNet-152, DenseNet-161, and Inception-v3), achieving significant improvement in FNAC image classification. For the imbalanced dataset problem, the number of positive samples is always far less than the negative ones in cytological scenarios (Yu et al., 2021; Bakht et al., 2020). A few studies employed sampling techniques to balance different classes (Li et al., 2019; Bakht et al., 2020). For learning-based solutions, Yu et al. (2021) adopted GAN to balance different classes by synthesizing images for those classes with far less amount.

Recently, some other advanced methods have been investigated for cytology classification. Li et al. (2022) introduced an attention mechanism block to guide the network to focus on cell areas, improving deep feature extraction. Then, they added a pyramid pooling layer and a long short-term memory module (LSTM) to aggregate image features in different regions. To improve the classification performance, Shi et al. (2021) proposed a cervical cell classification model based on graph convolutional network (GCN), which can explore the potential relationship between cervical cells.

In the clinic, DL-based classification approaches have been widely applied for various types of organs, including cervix (Shanthi et al., 2019), breast (Miselis et al., 2019), lung (Teramoto et al., 2017), thyroid (Bakht et al., 2020), urine (Vaickus et al., 2019), nasal (Dimauro et al., 2019) and skin (Noyan et al., 2020). Specifically, Vaickus et al. (2019) proposed a VGG-based model for classifying urine cytopathology images. Bakht et al. (2020) developed a VGG-based model for thyroid nodule cell classification. For skin cytology, Noyan et al. (2020) proposed a ResNet-based model to identify cells in the cytology of erosive-vesiculobullous diseases. These studies demonstrated the substantial clinical value of classification-assisted cytology image analysis.

4.2.2. Slide-level classification

Unlike cell-level classification, which focuses on identifying the type of each cell, the slide-level classification aims to grade the malignancy risk of whole slides. Several properties of whole slide data, such as giga-scale slides, sparse positive samples, multi-scale resolution, and labeling difficulty make this task extremely challenging.

Giga-pixel WSI classification systems are being investigated for efficient and high-accuracy predictions. As illustrated in Table 1 (slide-level), most studies built slide-level classification systems with multi-stage designs. For example, Cheng et al. (2021) designed a robust and progressive WSI analysis model for cervical cancer screening. The authors first developed a progressive lesion cell recognition method by combining low- and high-resolution WSIs. Subsequently, an RNN-based slide classification model was built for WSI-level predictions in the second stage. In another slide-level study, Wei et al. (2021) built a cervical WSI result prediction framework based on multi-scale semantic and spatial features. In the first stage of the framework, they designed a lightweight model (YOLCO) based on YOLO series (Redmon et al., 2016) to make local predictions (e.g., cell-level and patch-level), which can enrich the multi-scale connectivity by additional supervision of spatial information. In the second stage, these local predictions were input into a Transformer architecture for classifying abnormal *vs.* normal WSIs. Experimental results showed that the framework presented a higher classification performance (AUC = 0.872) with a speed of 538.01 s/slide, which is 2.51× faster than the state-of-the-art method, YOLOv3 (Redmon and Farhadi, 2018). For accurate and efficient screening of cervical cancer, Zhu et al. (2021) developed a complete cervical LBC smear TBS diagnostic system. The system integrated XGBoost and a logical decision tree with three typical DL models, i.e., Xception for classification, YOLOv3 for object detection, and U-Net for segmentation. The diagnostic system could reduce the overburdened workload on cytologists and improve the accuracy of cervical cancer screening.

Weakly supervised learning strategies are introduced to learn information from limited annotations, which is appealing in slide-level classification. For example, Dov et al. (2021) presented a MIL framework to predict thyroid cancer malignancy from cytopathology images. Then, an attention module was integrated into the MIL-based model with maximum likelihood estimation architecture. The experimental results showed competitive performance in thyroid malignancy prediction. Li et al. (2021) developed mixed supervision learning for WSI classification by effectively utilizing their various labels (e.g., sufficient image-level coarse annotations and a few pixel-level fine labels).

Many cytology solutions have investigated advanced strategies or designs for further classification performance improvements. For example, Cao et al. (2021) integrated the attention module into a multi-scale region-based CNN (feature pyramid network, FPN) between upsampling and downsampling pathway. Three experiments at different levels consisting of cell-level detection, patch-level, and case-level classification demonstrated the effectiveness of this strategy. Other studies designed different auxiliary tasks (e.g., detection and segmentation) to assist the classification task. Tan et al. (2021) employed an object detection model (Faster R-CNN) to integrate cervical cell detection and classification tasks into the same framework. This approach could provide the categories of detected cells instead of whole images containing multiple types of cervical cells. Ke et al. (2021) introduced U-Net as the segmentation network to output semantic predictions, such as the boundary of cytoplasm and nuclei. Then, only image patches with nuclei in the center can be fed into the classification network, improving the representation ability of the network for cell patches. Essentially, these auxiliary tasks are utilized to extract cell patches or semantic masks for improving the feature representation of cells by classification models.

Risk stratification is a crucial application of cytology classification, which determines the risk level of patients suffering from diseases. Awan et al. (2021) designed a DL-based risk stratification system for urine cytology. In this system, RetinaNet was employed for cell-level classification and detection in the first stage. Then, for slide-level risk stratification, they identified low-risk and high-risk cases using the count of atypical cells and the total count of atypical and malignant cells. Lin et al. (2021) presented a dual-path network for cervix risk stratification, which can be divided into two steps. Firstly, an efficient CNN with a dual-path encoder was proposed for lesion retrieval, which can ensure inference efficiency and sensitivity on both tiny and large lesions. Then, a smear-level classifier (rule-based risk stratification) was introduced to align reasonably with the intricate cytological definition of the classes. Extensive experiments on a large-scale dataset consisting of 19,303 WSIs from multiple medical centers validated the robustness of the risk stratification method.

4.3. Detection

In cytology screening, finding tiny objects (e.g., malignant cells and nuclei) could be extremely tedious and time-consuming for cytopathologists. Therefore, developing automatic detection models to search for and localize these objects could be essential for cytology image analysis, especially for whole slide analysis, as this is a prerequisite for downstream tasks such as slide-level diagnosis. Typically, accuracy and efficiency are two main considerations for detection methods in cytology screening systems. In DL-based object detectors, one-stage detectors have shown the superiority of detection efficiency in multiple object detection scenarios, while two-stage detectors offer better performance in accurate localization (see Table 4).

Table 4
Overview of deep learning-based detection studies for computational cytology.

Reference	Application	Staining	Organ	Method	Dataset	Result
One-stage methods						
Kilic et al. (2019)	Nuclei detection	Pap	Pleural effusion	YOLOv3	Private dataset: 200 images with 11,157 nuclei	Precision: 0.941, Recall = 0.9898, F1 = 0.9648, Test time = 0.060 s/img (image size is 416 × 416).
Xiang et al. (2020)	Automation-assisted cervical cancer reading	Feulgen	Cervix	YOLOv3	Private dataset: 12,909 images with 58,995 ground truth boxes in 10 categories	Detection: mAP = 0.602; Classification: Sens = 0.975, Spec = 0.687.
Tayebi et al. (2021)	Hematological diagnosis	Giemsa	Bone marrow	YOLOv4	Private dataset: 75,000 annotated tiles of bone marrow aspirate	Region detection: Acc = 0.97, AUC = 0.99; Cell detection: mAP = 0.75, F1-score = 0.78.
Liang et al. (2021a)	Cell detection	Pap	Cervix	YOLOv3 + Darknet-53 + Soft scale anchor matching	Private dataset: 12,909 cervical images with 58,995 ground truth boxes in 10 categories	mAP = 0.6544.
Two-stage methods						
Li et al. (2019)	Cell detection and classification	Pap	Cervix	Faster R-CNN + Transfer learning	Private dataset: 680 LBC cervical exfoliated cell samples	Classification: Acc = 0.9161 Detection: mAP = 0.6698.
Hossain et al. (2019)	Nuclei detection, Estimate proliferation rate	H&E	Kidney	R-CNN	Private dataset: 16,905 segmented cancer cell and 22,948 normal cell nuclei	P = 0.9901, R = 0.9870, F1 = 0.988.
Pirovano et al. (2021)	Localization and detection of abnormalities	Pap	Cervix	Weakly supervised CNN + Regression constraint	Herlev	Severity classification: Acc = 0.952; Normal/abnormal classification: Acc = 0.952, KAPPA score = 0.870; Detection: Acc = 0.804.
Chai et al. (2022b)	Cell detection	Pap	Cervix	Faster R-CNN + Deep metric learning	Private dataset: 240,860 images	75% labeled: mAP = 0.254; 50% labeled: mAP = 0.195.
Su et al. (2020)	Ascites cytopathology interpretation	Pap, H&E	Stomach	Classification: pre-trained AlexNet, VGG-16, GooleNet, ResNet-18, ResNet-50. Detection: Faster R-CNN	Ascites 2020	Classification: AUC = 88.51 (ResNet-50); Detection: IoU = 0.8722, mAP = 0.8316.
Xie et al. (2018)	Cell detection	H&E	Cervix	Fully residual CNN + Structured regression	HeLa cervical cancer	Precision = 0.98, Recall = 0.98, F1 = 0.98.
Marzahl et al. (2020)	Quantification of pulmonary hemosiderophages	Prussian turnbull	Lung	RetinaNet, FPN	Private dataset: 17 WSIs with 78,047 hemosiderophages	Concordance = 0.85, mAP = 0.66.
Zhang et al. (2019)	Cervical cytology analysis	Pap	Cervix	Cell detection: Faster R-CNN and RetinaNet. Classification: Inception-v3, ResNet-101, DenseNet-121	Private dataset: 1167 WSIs with 14,432 image patches, and 27,972 labeled lesion cells	Detection: mAP = 0.2116 (Faster R-CNN); Classification: Acc = 0.8884, F1 = 0.5996 (DenseNet-121).
Liang et al. (2021b)	Cell/clumps detection	Pap	Cervix	Faster R-CNN + Prototype + Few-shot learning	CDetector	mAP = 0.488.
Baykal et al. (2020)	Nuclei detection	Pap	Pleural effusion	Detector: Faster R-CNN, R-FCN, SSD	Private dataset: 200 images (11,157 nuclei)	Faster R-CNN (ResNet-101): F1 = 0.9812.

4.3.1. One-stage methods

One-stage algorithms detect objects by directly generating the category and coordinates of objects with the advantages of high detection efficiency, such as SSD (Liu et al., 2016), YOLO (Redmon et al., 2016), and RetinaNet (Lin et al., 2017b).

Many solutions in cytological object detection introduce the YOLO model as their base network due to its high efficiency. YOLO, a typical one-stage object detector, divides the original image into $S \times S$ grid cells. Then, YOLO predicts bounding boxes and confidence scores for each cell. Afterwards, redundant boxes are removed by the confidence threshold and non-maximum suppression. Xiang et al. (2020) used YOLO as their detector for cervical cells. Similarly, Kilic et al. (2019) adopted YOLO to detect nuclei in pleural effusion cytology. In the experiments, the authors compared the detection efficiency between one-stage and two-stage detectors. The results showed that YOLO achieved a test speed of 0.060 s/image, which was much faster than 1.627 s/image in Faster R-CNN (two-stage detector). Besides, Tayebi et al. (2021) applied YOLO on selected appropriate ROI tiles to automatically detect and classify bone marrow cellular and non-cellular objects. To improve the performance of YOLO in cervical cell detection, Liang et al. (2021a) proposed a global context-aware framework by introducing an image-level classification branch and a weighted loss that can filter false positive predictions.

For exploiting multi-scale features, RetinaNet was proposed by adopting FPN as its feature extractor, which achieved the state-of-the-art detection performance (Lin et al., 2017a). Marzahl et al. (2020) employed RetinaNet to generate rich and multi-scale features for functional heads (e.g., box, regression, and classification). These results contributed to the quantification of pulmonary hemosiderophages in this work.

4.3.2. Two-stage methods

Two-stage detectors typically follow the region proposal paradigm with the first stage generating bounding boxes, followed by box regression and classification. Typical region proposal strategies include sliding windows (Girshick, 2015), selective search (Van de Sande et al., 2011), and RPN (Ren et al., 2015). Specifically, selective search strategy is designed in Fast R-CNN to generate bounding boxes. Then, the ROI pooling layer extracts the features of each ROI. Fast R-CNN outputs softmax probabilities and per-class bounding-box regression offsets with a multi-task loss (Girshick, 2015). To integrate different modules and increase the detection speed (Ren et al., 2015), Faster R-CNN improves Fast R-CNN by integrating feature extraction, region proposal, bounding box regression, and classification. For further acceleration, RPN for region proposal is designed in Faster R-CNN to replace selective search strategy in Fast R-CNN.

To detect cellular objects in whole cytology images, some studies employed Faster R-CNN as their base architecture. For example, Li et al. (2019) utilized Faster R-CNN to detect cervical exfoliated cells from LBC images. Similarly, Faster R-CNN was also used to detect tumor cells for further classification, which formed an ascites cytopathology image interpretation system (Su et al., 2020).

For different cytological scenarios, researchers improved detection performance by modifying architectures or integrating with other strategies (Hossain et al., 2019; Marzahl et al., 2020). For efficient cell and robust detection, Xie et al. (2018) presented a structured regression model based on a proposed fully residual CNN. This model produced a dense proximity map to exhibit higher responses at the cell centers. Then, training this model only required annotations of dots instead of conventional boxes, which can improve the annotation efficiency. Several studies investigated weakly supervised learning settings in cytological detection. For example, Pirovano et al. (2021) proposed a computer-aided diagnosis tool for cervical cancer screening. In this research, the authors designed a weakly supervised localization strategy, which performed the Integrated Gradient method (Sundararajan et al., 2017) to compute attribution maps and morphological operations to obtain the localization boxes. In another work, Chai et al. (2022b) proposed a semi-supervised deep metric learning method to improve intra-class feature compactness for cervical cancer cell detection. This study proposed to exploit an embedding metric space and dual aligned semantic features on proposal and prototype levels. Their quantitative experiments showed that detecting cervical cancer cells can be challenging, especially for rare types of cancers, such as ASC-US.

The clinical practice requires not only high detection accuracy but also efficiency as detection speed is the main concern in large-scale screening scenarios (Lin et al., 2019a). Detection models usually struggle with the trade-off between accuracy and efficiency. For example, two-stage detectors (e.g., Faster R-CNN) can achieve higher localization accuracy while one-stage detectors (e.g. YOLO) have advantages in faster detection speed. Zhang et al. (2019) compared performances of one-stage (RetinaNet) and two-stage (Faster R-CNN) methods for the detection of cervical lesion cells in cytological studies. The results showed that the former achieved better experimental results in average precision. In another work (Liang et al., 2021b), the authors improved Faster R-CNN and compared the performance with the baseline and RetinaNet in a limited data scenario. The cervical cell/clump detection results showed that RetinaNet (one-stage) achieved faster speed (FPS). Baykal et al. (2020) compared three detectors, i.e., Faster R-CNN (two-stage), R-FCN (two-stage), and SSD (one-stage). The results showed that R-FCN achieved a higher mAP score while SSD spent less time on the test set. These experimental results demonstrated the trade-off of detection models between efficiency and accuracy.

4.4. Segmentation

In cytology image analysis, the segmentation task could be used to (1) separate cells/clump and background from specimens, (2) morphologically distinguish cell types, and (3) accurately segment cellular structures (e.g., nucleus and cytoplasm).

Accurately segmenting overlapping areas between cells could be one of the most critical challenges in cytology image analysis with two public datasets (ISBI2014, ISBI2015) focusing on this challenge (Lu et al., 2015, 2016). There are two main solutions to address this issue. One is the bottom-up approach which divides the cytological segmentation task into two stages. The first stage utilizes a semantic segmentation model (e.g., U-Net) for obtaining coarse predictions, followed by a series of refinement designs for overlapping areas, obtaining the final accurate segmentation result. The other is based on the top-down paradigm (e.g., Mask R-CNN He et al., 2017), which detects cytology objects in whole images and outputs a segmentation map by mask prediction head. Therefore, we divide the solutions of cytology segmentation into two categories: bottom-up approach and top-down approach (see Table 5).

4.4.1. Bottom-up approach

DL-based models such as U-Net provide solutions to segment cellular objects in cytology images. Then, segmenting each cell can be hampered by overlapping areas belonging to multiple cells. To overcome this issue, various post-processing strategies have been designed for fine-level segmentation.

The segmentation network was originally implemented by establishing a CNN-based pixel-wise classification network. Song et al. (2015) designed a multi-scale convolutional network for coarse segmentation. Kowal et al. (2020) proposed a more complex CNN structure consisting of four convolutional layers, two max-pooling layers, and one fully connected layer. This architecture was utilized in the first stage for nuclei segmentation from cytological images.

U-Net almost replaces the previous pixel-level classification network after its occurrence, especially in biomedical image segmentation (Ronneberger et al., 2015). U-Net adopts the downsampling–upsampling architecture which provides end-to-end semantic segmentation. Specifically, two sampling parts are respectively used to learn feature maps in a multi-resolution manner, and the extracted feature maps are restored to the same size as the input image. Then, skip connection and feature map concatenation strategies are introduced to combine low-level and high-level features. Recently, U-Net has also made great achievements in cytology image segmentation. For example, Falk et al. (2019) designed U-Net for cell counting, detection, and segmentation. This work illustrated its potential for cellular structure analysis. In cytology image segmentation, U-Net has been introduced as the backbone for segmenting cellular objects from different organs, such as oral (Matias et al., 2021), cervix (Araujo et al., 2019), and breast (Kowal et al., 2020). Several studies focus on improving the performance of U-Net to enhance the model capacities for cell segmentation. For instance, Böhm et al. (2019) proposed to mix 2D and 3D U-Net for semantic instance segmentation of touching objects. Zhang et al. (2020) introduced the attention mechanism, enabling U-Net to focus on ROIs. Besides, Hussain et al. (2020b) improved U-Net by adding residual blocks, densely connected blocks, and a fully convolutional layer as a bottleneck between encoder–decoder blocks for nuclei segmentation in cervical images. These studies adopted U-Net to predict pixel-level semantic segmentation results of cytology images, achieving satisfactory performance in non-overlapping objects. For overlapping regions, such methods require further refinement designs because the pixels in these regions need to be predicted as the foreground of multiple instances, while the conventional U-Net only outputs the probability of categories.

In bottom-up strategy for cytology segmentation, the second stage is to handle overlapping regions and refine segmentation results. Most solutions take task-specific knowledge (e.g., cell shape prior) into consideration. For instance, Song et al. (2016) proposed a dynamic multi-template deformation model together with high-level morphological constrains for further boundary refinement. Kowal et al. (2020) designed a series of refinement strategies in the second stage: conditional erosion for determining nuclei seeds, the seeded watershed for separation of overlapping nuclei, and aggregating segmentation results for

Table 5
Overview of deep learning-based segmentation studies for computational cytology.

Reference	Application	Staining	Organ	Method	Dataset	Result
Bottom-up approach						
Falk et al. (2019)	Cell counting, detection, and morphometry	Fluore-scence	Various	U-Net	ISBI cell tracking 2015	IoU = 0.9203 (PhC-U373), IoU = 0.7756 (DIC-HeLa).
Matias et al. (2021)	Segmentation, detection, and classification of cell nuclei	Pap	Oral	Classification: ResNet-34. Detection: Faster R-CNN. Segmentation: U-Net	Oral 2021	Classification: Acc = 0.88, F1 = 0.86; Detection: IoU = 0.5832; Segmentation: IoU = 0.4607.
Song et al. (2014)	Segmentation of cytoplasm and nuclei	H&E	Cervix	CNN+ Coarse to fine segmentation	Private dataset: 53 slides with 1400 cells	Nuclei detection: F1 = 0.9453; Segmentation: F1 = 0.8951.
Song et al. (2015)	Segmentation of cytoplasm and Nuclei	H&E	Cervix	Multi-scale + Graph partitioning + Touching cell splitting	Private dataset: 53 images (slide)	Cytoplasm: Dice = 0.95; Nuclei: Dice = 0.99.
Song et al. (2016)	Cell segmentation	Pap H&E	Cervix	Multi-scale CNN + Dynamic multi-template deformation	ISBI 2015. Private dataset: 21 images (each image has 30~80 cells)	ISBI 2015: Dice = 0.89; Private dataset: Dice = 0.84.
Araujo et al. (2019)	Cell image segmentation and ranking	Pap	Cervix	CNN	BHS 2019	Segmentation: P = 0.73, R = 0.65, F1 = 0.69, Time = 4.75s; Ranking: mAP = 0.936.
Kowal et al. (2020)	Cell nuclei segmentation	H&E	Breast	CNN + Seeded watershed	Public dataset: 80 images	Benign: Jaccard distance = 0.776; Malignant: Jaccard distance = 0.732.
Böhm et al. (2019)	Semantic instance segmentation of touching and overlapping objects	Pap	Cervix	U-Net	OSC-ISBI	Dice = 0.895 ± .0.079.
Zhang et al. (2020)	Cervical cell segmentation	Pap	Cervix	Attention mechanism + U-Net + Random walk	ISBI 2014	Nuclei: P_p = 0.94, R_p = 0.95, Dice = 0.93; Cytoplasm: Dice = 0.93.
Walter et al. (2021)	Instance segmentation	Pap	Cervix	U-Net + Star-convex polygons	OSC-ISBI	Dice = 0.85 ± 0.07.
Hussain et al. (2020b)	Segmentation and classification of cervical nuclei	Pap	Cervix	U-Net	Herlev	Classification: Acc = 0.988; Segmentation: ZSI = 0.97.
Tareef et al. (2017)	Cytological examination (overlapping cell segmentation)	Pap	Cervix	CNN+ Shape prior (dynamic shape modeling)	ISBI 2014	Nuclei: P_p = 0.94 ± 0.06, R_p = 0.95 ± 0.06, ZSI = 0.94 ± 0.04; Cytoplasm: ZSI = 0.90 ±0.08.
Song et al. (2020)	Overlapping cytoplasms segmentation	Pap H&E	Cervix	Shape mask generator + Refining shape priors	ISBI 2015. Private dataset: 160 clumps with 962 cytoplasms	Pap: Dice = 0.854 ± 0.049; H&E: Dice = 0.846 ± 0.054.
Wan et al. (2019)	Nuclei detection, cytoplasm segmentation	Pap	Cervix	CNN + Double-window + Deeplab v2 + CRFs + Cell boundary refinement	ISBI 2014; ISBI 2015; Private dataset: 580 image (patch)	ISBI 2014: Dice = 0.93 ± 0.04; ISBI 2015: Dice = 0.92 ± 0.05; Private: Dice = 0.92 ± 0.04.
Top-down approach						
Sompa-wong et al. (2019)	Pap smear cancer screening	Pap	Cervix	Mask R-CNN + Fine-tuning	Private dataset: 178 images in classes of normal (2734), atypical (494), low-grade (148), high-grade cells (84)	Image-level: mAP = 0.578, Acc = 0.917; Nucleus: Acc = 0.898, Sens = 0.725, Spec = 0.943.
Zhou et al. (2019b)	Cell segmentation	Pap	Cervix	PRN + Cell association matrix	Private dataset: 413 images (annotated 4439 cytoplasm and 4789 nuclei)	Cytoplasm: AJI = 0.7185, F1 = 0.7497; Nuclei: AJI = 0.5496, F1 = 0.7554.
Zhou et al. (2020)	Cell instance segmentation	Pap	Cervix	RPN + Knowledge distillation	Private dataset: 413 labeled (4439 cytoplasm and 4789 nuclei) and 4371 unlabeled images	80%. labeled: AJI = 0.6692, mAP = 0.4013; 40% labeled: AJI = 0.6449, mAP = 0.3726.

overlapping and non-overlapping nuclei. In another research, Zhang et al. (2020) proposed a graph-based random walk method to extract both nucleus and cytoplasm of overlapping cervical cells. This approach utilized the polar coordinate sampling strategy to remove false positive predictions that are much smaller in size than normal nuclei. The experimental results in ISBI 2014 dataset showed the improvement in separating individual cells from heavy overlapping cell clumps. Walter et al. (2021) proposed to predict object probability, star distance, and overlap probability based on U-Net. Then, non-maximum suppression was used to generate overlapping cell segmentation results. These refinement strategies can achieve more accurate segmentation results,

Table 6
Overview of deep learning-based studies of other tasks for computational cytology.

Reference	Application	Staining	Organ	Method	Dataset	Result
Ma et al. (2020)	Super resolution	Pap	Cervix	Image registration + GAN	Private dataset: 142 WSIs (118 for training and 24 for testing) with 174,500 patches	PSNR = 26.92, SSIM = 0.88, MOS = 3.80.
Ma et al. (2021)	Super resolution	Pap	Cervix	Backbone + Self-texture + Flexible reconstruction	Public dataset: 5 slides (25,000 patches)	PSNR = 35.47, SSIM = 0.958, MSE = 22.07.
Teramoto et al. (2021)	Mutual stain conversion	Giemsa and Pap	Lung	CycleGAN	Private dataset: 191 Giemsa-stained images and 209 Pap-stained images	T test: *P*-value <0.001.

especially for overlapping regions. However, complex clinical data will present more challenges to reproducibility and generalizability.

In addition, some other models have been proposed for cytological segmentation. Tareef et al. (2017) designed a two-stage segmentation model, which consists of initial segmentation based on the Voronoi diagram, and final segmentation with learning shape prior model. In order to segment overlapping cervical cytoplasms, Song et al. (2020) proposed a shape mask generator to refine shape priors. Wan et al. (2019) presented an architecture for cell detection and cytoplasm segmentation. This approach utilized conditional random field algorithms (CRFs) and cell boundary refinement to achieve accurate segmentation of overlapping cells in cervical cytology.

4.4.2. Top-down approach

The top-down paradigm for segmenting instance objects involves two steps within a single framework: generating all ROIs in the whole image and then segmenting instances from each detected ROI. As a popular architecture of this paradigm, Mask R-CNN (He et al., 2017) improves Faster R-CNN (Ren et al., 2015) by adding full connected layers as the mask prediction head. Thus, it can output the prediction of classification, detection, and segmentation via a single architecture (Ren et al., 2015).

As one of the most challenging tasks in cytology image analysis, instance segmentation involves predicting the instance morphology and distinguishing different instances (e.g., cytoplasm and nucleus). Top-down models could be helpful to address this challenge since they can segment objects from each detected instance using an end-to-end architecture without any complex refinement design. A few studies investigated this category of cytological segmentation methods. Due to its success in instance segmentation, existing researches typically take Mask R-CNN as their architecture (He et al., 2017). For example, Sompawong et al. (2019) adopted Mask R-CNN to specify the bounding box, nucleus mask, and class of each cervical cell. In another study (Zhou et al., 2019b), authors utilized the multi-head attention mechanism to explore instance-level association by propagating features based on attention scores. Consequently, the proposed model improved the instance representation and achieved better instance segmentation performance than the original Mask R-CNN. To further reduce reliance on large amounts of labeled data, Zhou et al. (2020) proposed a semi-supervised learning approach to leverage both labeled and unlabeled data for instance segmentation by knowledge distillation.

4.5. Other tasks

In addition to typical DL tasks, i.e., classification, detection, and segmentation, a few other tasks of cytology image analysis have also been investigated, such as super-resolution (SR), and stain conversion (Table 6). Cytology images that are out-of-focus and low-resolution will impede the decision-making process of cytologists during cytopathology screening. Hence, high-resolution digital cytopathology slides are the prerequisite for the interpretation of lesion cells. To control the image quality, super-resolution models are designed to generate high-resolution images. Ma et al. (2020) introduced a GAN-based progressive multi-supervised super-resolution model (PathSRGAN) to learn the mapping between low-resolution and high-resolution images. After that, they designed a self-texture transfer super-resolution and refocusing network (STSRNet) to reconstruct HR multi-focal plane (MFP) images from a single 2D low-resolution (LR) wide field image (Ma et al., 2021). As mentioned in Section 4.1, different staining techniques are used to enhance the visual appearance of cellular structures and components. DL-based stain conversion can be used for staining normalization and mitigating data heterogeneity issues. Teramoto et al. (2021) proposed a CycleGAN-based style transfer model for stain conversion between Giemsa-stained and Pap-stained images. This study performed visual evaluations of the authenticity of cell nuclei, cytoplasm, and cell layouts of synthetic lung cytology images.

5. Challenges and promises

Despite great advancements and improvements in computational cytology over the last few years, there are still quite a few challenges and opening problems waiting to be resolved. Meanwhile, the development of DL technologies and cytology is continuously accelerating this emerging field. This section further discusses prospects and potential research directions of computational cytology.

5.1. Label-efficient learning with limited annotations

Data is regarded as a vital prerequisite for learning-based methods since it is difficult to develop robust and effective models without high-quality annotated datasets (Falk et al., 2019; Jónsson et al., 2019). In medical image analysis, large-scale labeling can be a heavy burden for cytologists. They need to manually delineate ROIs in WSIs, then annotate each object (e.g., nucleus, cell, and cluster) by a box or mask. Compared with the large-scale datasets in histopathology (e.g., TCGA Tomczak et al., 2015), public cytology datasets are more limited not only in their numbers but also in cancer types and annotation types (shown in Table 1). Therefore, how to efficiently utilize datasets with limited annotations can be challenging for developing DL-based cytology image analysis models (Wang et al., 2019b; Chai et al., 2022b).

In recent years, label-efficient learning has been proposed to make full use of limited annotations for leveraging information, including semi-supervised learning (Wang et al., 2021), multiple instance learning (Wang et al., 2019a), mixed supervised learning (Li et al., 2021), etc. Specifically, semi-supervised learning aims to solve this problem by learning knowledge from both labeled data and unlabeled data. Recently, Zhou et al. (2020) designed a semi-supervised learning method using a teacher–student framework for overlapping cell instance segmentation. Another solution, MIL utilizes image-level annotations for instance-level tasks, which has been investigated in the field of medical image analysis, especially for histopathology. Dov et al. (2021) proposed a MIL-based algorithm in thyroid cytology, which can simultaneously predict multiple bags and instance-level labels for thyroid malignancy prediction from WSIs. However, the potential of MIL in pap smears and other cytology images remains to be explored. As illustrated in Table 1, the annotation type usually varies in different datasets, including box, mask, and class label. Currently, mixed

supervised learning has gained popularity in analyzing images with different types of annotations. It has been preliminarily demonstrated as an effective learning scheme in medical image analysis. For example, Luo et al. (2021) presents a deep omni-supervised thoracic disease detection network from chest X-rays with massive image-level annotations and scarce lesion-level annotations. This omni-supervised learning paradigm can substantially reduce the demand for fine annotation, reducing the workload of doctors significantly (Chai et al., 2022a).

In addition to the effective utilization of annotations, it is also promising to develop efficient labeling approaches to reduce the burden on annotators. For example, introducing human knowledge into the annotation loop can effectively and actively obtain accurate and credible annotations. Koohbanani et al. (2020) proposed an annotating method named NuClick with a point or squiggle as a guiding signal, enabling the segmentation of cell boundaries. However, this method still requires full attention of human, which is a huge cost for society and tedious for human experts. Since cytology images always contain numerous cells, especially in giga-pixel WSIs, how to establish an effective labeling process remains largely unexplored.

5.2. Fine-grained classification and morphological feature characterization

In spite of the numerous DL applications developed in cytology image analysis (e.g., classification, detection, and segmentation), some of these tasks deserve further investigation, such as fine-grained classification for cancer screening and instance segmentation in overlapping cell scenarios.

As a fundamental task, cytological classification aims to distinguish between benign and malignant cells. Clinically, there are many subtypes of cells or lesions with varying degrees of cancerization. For example, the interpretation results of cervical liquid-based cytology smears comprise four main types and each type contains several subtypes: normal, squamous lesions (ASC-US, LSIL, ASC-H, HSIL, SCC), glandular lesions (AGC-NOS, AGC-FN, AIS, ADC), and infectious lesions (TRI, CAN, HSV, ACTINO, CC) (Nayar and Wilbur, 2015). Inter- and intra-variances among different types and subtypes present significant challenges for DL-based feature extractors to exploit distinguishable features. Further fine-grained classification can solve these problems and assist cytologists in accurate cancer diagnosis. Zhang et al. (2019) demonstrated the difficulty of fine-grained tasks compared to coarse-grained tasks. They found that the classifier of cervical cytology performed significantly worse on fine-grained tasks than on coarse-grained tasks. Additional information such as prior knowledge could be helpful to learn underlying and discriminative feature representations. For example, as morphological information, the masks of cytological objects (e.g., cytoplasm and nucleus) have been demonstrated the effectiveness to assist the fine-grained classification task (Lin et al., 2019b). Therefore, fine-grained feature extractors for learning subtle feature diversity are essential for cytology image analysis.

Another typical cytological application, instance-level segmentation of cytoplasm and nucleus can be used to calculate the nuclear-cytoplasmic ratio, which is an essential indicator for distinguishing malignant cells. However, cell overlapping regions pose challenges to accurately segment cellular structures (Song et al., 2014, 2015). Most approaches address this challenge dividing the task into two stages: coarse segmentation via semantic segmentation models, followed by refinement processing technologies for overlapping regions (Song et al., 2016; Zhang et al., 2020). These multiple-stage approaches introduce human designs and interventions, causing training difficulties and poor generalization. Recently, researchers have investigated the feasibility of building end-to-end models using the top-down segmentation paradigm (Zhou et al., 2019b), which can detect objects and then segment each detected object via a single framework. Even though segmentation task has made some progress in cytology image analysis, building end-to-end segmentation models is still valuable to explore for complex morphological feature representation.

5.3. Effective feature representation learning

In DL-based models, downstream tasks will be greatly affected by the feature representation capability of feature extractors. The main goal of feature extractor in computational cytology is to learn effective features of cytology images for cell classification, cellular object detection, and segmentation. Thus, building models that can effectively represent features is essential for cytology image analysis.

Recently, the attention mechanism has made great progress in effectively representing image features, improving performance of downstream tasks (Vaswani et al., 2017). Each cytological image typically contains quite a few useless objects, such as background, mucus, blood, and inflammatory cells (Hussain et al., 2020a). Attention mechanism-based strategies can make DL models focus more on lesion-related areas or object-related (e.g., nuclei and cytoplasm) regions. Besides, the attention mechanism can not only improve the performance of DL model but also provide convenience for model visualization. Several studies have validated the superiority of the attention mechanism in cytological tasks. Zhang et al. (2020) utilized the attention mechanism to enhance U-Net for segmentation of overlapping cervical cells. Besides, Zhou et al. (2019b) introduced a multi-head attention mechanism module into the instance segmentation model to improve instance representation. These studies integrated the attention modules into the DL framework and achieved improved performances. Recently, a popular structure with an attention mechanism, Transformer, has then been demonstrated to be superior in learning global dependencies in various applications (Dosovitskiy et al., 2020). The superiorities of Transformer-based structures still remain to be evaluated in cytological applications, such as capturing long-range dependencies between different objects in patch images, or different regions in WSIs.

5.4. Generalizability and robustness

The generalizability of DL-based algorithms in computational cytology determines whether the model can be successfully applied to actual clinical scenarios with various influences. At present, most DL approaches for medical image analysis can obtain acceptable performance in their in-house datasets, but their performance in clinical settings falls well short of practical standards (Litjens et al., 2017).

The heterogeneity from specimen collection methods, staining techniques, and imaging protocols can significantly impact the robustness and generalization capabilities of DL models. Clinically, performances of these DL models may degrade significantly when applied to unseen data, leading to low clinical reproducibility (Beam et al., 2020). Furthermore, due to the discordance among experts, the variability of inter- and intra-observer causes the model to perform differently in specific annotated sets and clinical sets (Lodha et al., 2008).

Extensive researches were conducted to mitigate the above problem (Zhou et al., 2021). Previous studies used normalization to reduce the appearance variation during data preprocessing. However, two datasets can be influenced and normalized against each other, the solution provides limited benefit in DL-based medical image analysis (Mahmood et al., 2019). For learning-based solutions, domain adaptation can alleviate this problem by transferring knowledge to decrease the need for annotations of target tasks (Oza et al., 2021). Domain adaptation has been primarily used in other medical scenarios with cross-domain data (e.g., CT and MRI Chen et al., 2020), and the potential of domain adaptation methods remains to be explored in cytological scenarios. For instance, building cross-domain learning frameworks to analyze multi-domain images, such as multi-center images, differently stained cytology images, and even different types of pathology images (e.g., cytology and histopathology).

Designing models for robust feature extraction can also be worthy of exploration, especially for the robust feature extractor, which aims to map cytology images from multiple domains to the same feature space for learning domain-invariant representations, thereby addressing the

data heterogeneity issue (Li et al., 2021). In addition, multimodal data has been demonstrated to compensate for missing information in single-modal data (Zhang et al., 2021). For early-stage cancer prognosis, Shao et al. (2019) designed a multimodal fusion framework to integrate information from histopathological images and genomic sequences. In computational cytology, multimodality and full modality learning are waiting to be unexplored.

5.5. Transparency and interpretability

In clinical decision-making workflows, it is essential to consider not only the model performance but also interpretability (Tjoa and Guan, 2020). However, the black-box nature of DL algorithms lacking clinical interpretability and transparency has restricted its clinical generality and usability (Lu et al., 2021). Therefore, explainable AI for medicine can establish trust between AI technologies and doctors. There are some explorable issues in computational cytology, such as slide-level cytology screening based on rules, and visualization of the DL decision-making process.

The visualization technique is typically regarded as a major solution for model interpretation. Currently, techniques rely on the attention mechanism to generate heatmaps for visualization, such as class activation mapping (CAM) (Zhou et al., 2016). In cytology image analysis, a few studies have employed heatmaps or feature maps for exploring the decision-making basis of DL models in the classification task. For example, Nojima et al. (2021) used Grad-CAM to observe high response regions in cytology images, which can improve medical decision-making from the gradient of the differentiable model. In terms of model-agnostic visualization, another technique, local interpretable model-agnostic explanation (LIME) (Ribeiro et al., 2016) can effectively provide interpretability and transparency. LIME presents a locally faithful explanation by fitting a set of perturbed samples near the target sample using a potentially interpretable model. More visualization techniques are under exploration for interpretability.

To increase the credibility of DL in computational cytology, researchers have attempted to construct slide-level cytology screening systems by compiling multiple tasks rather than predicting the final diagnosis results, which can provide cytologists with multi-stage analysis for obtaining the final screening result. Zhu et al. (2021) proposed to integrate classification, detection, and segmentation to build a cervical LBC smear TBS diagnostic system. Another study, Lin et al. (2021) built a dual-path network for outputting the detected lesions, followed by a rule-based risk stratification system. Wei et al. (2021) divided the cervical WSI analysis into two stages, i.e., cervical lesion detection at the patch level in preselected ROIs, and normal/abnormal classification at the WSI level. The output of each stage of these multi-stage methods can provide intermediate explanations for model prediction and cytology screening, improving the transparency of diagnostic workflows.

5.6. Digital medicine and human-AI collaboration

Digital medicine is to integrate medicine and information technology for clinical diagnosis and treatment (Samuel et al., 2020). Although a large number of DL models report that they can achieve state-of-the-art performance, they are mostly validated on domain-specific datasets. They can hardly obtain clinically satisfactory performance. Data unavailability is one of the most crucial factors leading to this problem, since data privacy cannot be overemphasized in the medical domain. Currently, privacy-preserving learning solutions such as federated learning (Rieke et al., 2020) are being explored to improve the availability and privacy of multi-center data (Chen et al., 2021a). Clinical data is usually more diverse and complex than training data, which is another essential aspect of clinical transformation. This is caused by factors including imaging microscopes, staining techniques, patch extraction, selection, etc. To address this issue, designing robust models can make it less dependent on data quality. Moreover, cytologists analyze specimens of different cancer types using different diagnostic criteria (Nayar and Wilbur, 2015), making it difficult for DL algorithms that focus on domain-specific data to adapt to different scenarios.

In recent years, many studies have shown that the human–machine collaborative medical diagnosis system can achieve better diagnosis performance than conventional diagnosis systems with the intervention of human experts and the assistance of machines (Zhu et al., 2021; Patel et al., 2019). DL algorithms in intelligent medical systems provide doctors with high-confidence, multi-level prediction results. Besides, the digital diagnostic system can provide doctors with real-time diagnostic information through human–computer interaction technology. A relevant pioneer effort, the smart microscope system (ARM) was designed by Google Health (Chen et al., 2019). ARM integrated AI algorithms with optical microscopes to image and analyze pathological slides, providing pathologists with multiple analysis results (e.g., lesion contour and probability heatmap) in the field of view by augmented reality technologies. Based on these quantitative and qualitative analysis, pathologists can make a final diagnosis. The diagnostic system has provided prospects for the automation of histopathology, cytology, parasitology, etc. However, there is still a long way to go in terms of accurate and efficient diagnosis, system integration, hardware resources, and AI algorithms.

6. Conclusion

In this survey, we considered deep learning techniques for computational cytology and presented an extensive overview of existing approaches for cytology image analysis. From these surveyed approaches, different learning paradigms (supervised, weakly supervised, and unsupervised learning) have been applied in different organ types (cervix, breast, lung, thyroid, bladder, etc.) to develop various tasks, such as classifying benign and malignant cells, slide-level risk stratification, locating cells in cytological smears, segmenting cellular components. There are some remaining open challenges discussed in this paper, including data utilization, learning strategies, network architectures, and clinical problems, which brings the potential for developing advanced algorithms to benefit the development of computational cytology.

CRediT authorship contribution statement

Hao Jiang: Conceptualization, Methodology, Writing – original draft, Visualization. **Yanning Zhou:** Conceptualization, Formal analysis, Writing – review & editing. **Yi Lin:** Conceptualization, Writing – review & editing, Investigation . **Ronald C.K. Chan:** Formal analysis, Writing – review & editing. **Jiang Liu:** Conceptualization, Investigation. **Hao Chen:** Conceptualization, Funding acquisition, Project administration, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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